

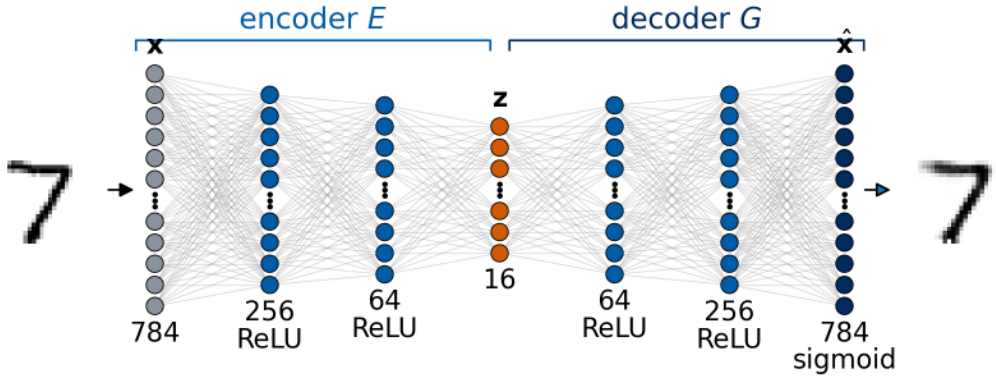
Lecture 6: Generative Models
Variational and Conditional Autoencoders
Generated Spins and the Hamiltonian

Deep Learning in Classical and Quantum Physics

Marcin Płodzień

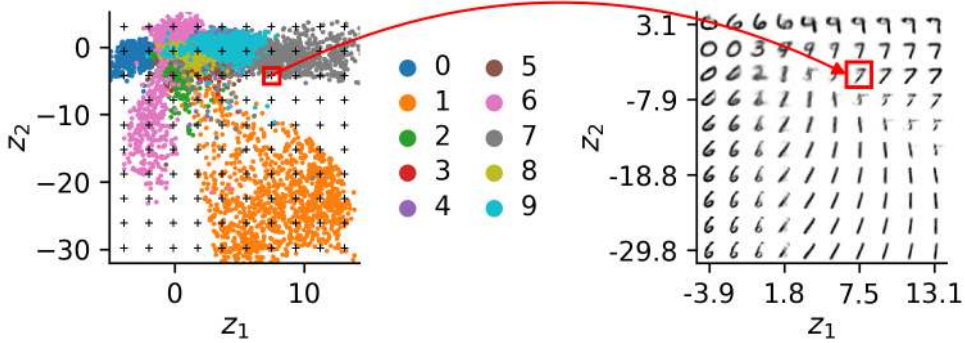
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The Autoencoder Compresses Each Image to a Point



Autoencoder (AE): $\mathbf{z} = E(\mathbf{x}) \in \mathbb{R}^d$, $d \ll 784$, $\hat{\mathbf{x}} = G(\mathbf{z})$; trained with the mean squared error $\frac{1}{n}\sum_i \frac{1}{784}\|\mathbf{x}_i - G(E(\mathbf{x}_i))\|^2$. Each image becomes *one point*; nothing is learned about latent vectors no image was mapped to.

Decoding a Grid of Latent Points



AE, $d = 2$; left: test codes $E(\mathbf{x})$, + grid; right: **cell at $(z_1, z_2) = G(\mathbf{z})$** (grey = pixel value).
Grid: equal steps, 1st to 99th percentile of the codes; red: a point and its image.

- neighbouring cells decode to similar images (G is continuous)
- one region per digit; between two regions the image morphs into the other digit
- the empty space between islands of codes decodes to non-digits (G never trained there)

Random Latent Vectors Give No Digits



AE with $d = 16$, decoder applied to random $\mathbf{z}$. Top: $\mathbf{z} \sim \mathcal{N}(0, I)$; bottom: $z_k \sim \mathcal{N}(m_k, s_k^2)$, with m_k, s_k the mean and standard deviation of $E(\mathbf{x})_k$ over the training set.

- the codes lie on a thin, curved set in $\mathbb{R}^{16}$; a random point almost never falls on it, even with the right mean and spread per coordinate
- a convolutional neural network (CNN) classifier (Section 4) is confident, $\max_c p(c \mid \mathbf{x}) > 0.9$, for 37% of such images, against 96% of real test digits

To generate we need a known distribution on the latent space, and a decoder trained on all of it.

1

Learning a Probability Distribution



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Generative Models

The data $\mathbf{x}_1, \ldots, \mathbf{x}_n$ are **samples** from an unknown probability distribution $p(\mathbf{x})$: handwritten digits, or spin configurations at temperature T, $p(\mathbf{s} \mid T) = e^{-H(\mathbf{s})/T}/Z$.

A **generative model**

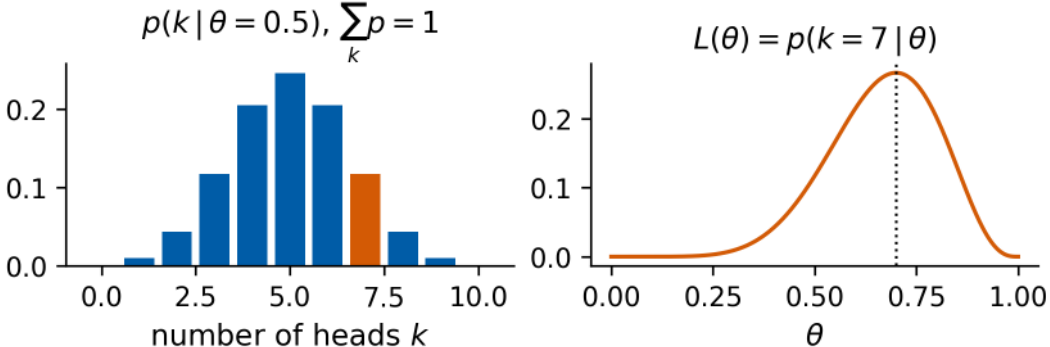
- learns an approximation $p_\theta(\mathbf{x}) \approx p(\mathbf{x})$ from the samples (θ: its parameters)
- draws **new samples** from it: instances that are not in the data set, but could have been

In science: new molecules and materials, ensembles of weather forecasts, fast surrogates of expensive simulations, and Monte Carlo samplers for lattice models.

Two steps: fit a distribution to data (maximum likelihood), and make it easy to sample (latent variables).

Probability and Likelihood

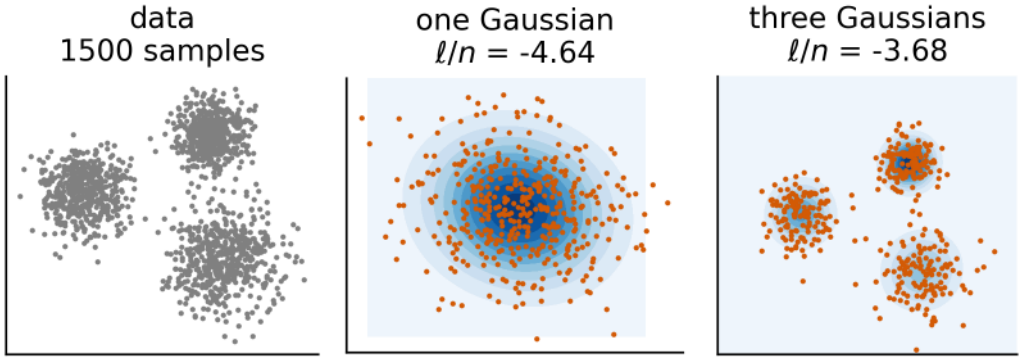
A model $p(x \mid \theta)$ is read in two ways: **probability** (θ fixed, x varies, $\sum_x p = 1$) and **likelihood** (data fixed, θ varies, $L(\theta) = p(x_{\text{obs}} \mid \theta)$, not a distribution over θ).



A coin tossed 10 times, $p(k \mid \theta) = \binom{10}{k}\theta^k(1-\theta)^{10-k}$. Left: $p(7 \mid 0.5) = 0.117$. Right: $k = 7$ observed, $\arg\max_\theta L = 0.7$, $\int_0^1 L\, d\theta = 1/11$.

Maximum Likelihood

Independent samples $\mathbf{x}_i$, model p_θ: $\ell(\theta) = \sum_{i=1}^{n} \log p_\theta(\mathbf{x}_i)$, $\theta^* = \arg\max_\theta \ell$. For $n \to \infty$, $\frac{1}{n}\ell \to -H(p) - \mathrm{KL}(p \,\|\, p_\theta)$, entropy $H(p) = -\mathbb{E}_p \log p$, **Kullback–Leibler divergence** $\mathrm{KL}(p\|p_\theta) = \mathbb{E}_p \log(p/p_\theta) \geq 0$ ($= 0$ iff $p_\theta = p$): $\max_\theta \ell \Leftrightarrow \min_\theta \mathrm{KL}(p \,\|\, p_\theta)$.



$p_\theta = \sum_k \pi_k \mathcal{N}(\mathbf{x}; \boldsymbol{\mu}_k, \Sigma_k)$, gradient ascent on ℓ; orange: 400 new samples.

Hierarchical Generation: Blueprint First, Microstate Second

Sampling high-dimensional data directly from $p(\mathbf{x})$ is intractable (e.g. 2^{900} spin states). Decompose generation into a **two-step hierarchical recipe**:

$$\mathbf{z} \sim p(\mathbf{z}) = \mathcal{N}(0, I) \text{ (macrostate)}, \qquad \mathbf{x} \sim p_\theta(\mathbf{x} \mid \mathbf{z}) \text{ (microstate via decoder)}$$

- **Law of total probability**: integrate over all possible hidden causes,

$$p_\theta(\mathbf{x}) = \int p(\mathbf{z})\, p_\theta(\mathbf{x} \mid \mathbf{z})\, d\mathbf{z}$$

- **Physical meaning**: to find the likelihood of microstate $\mathbf{x}$, integrate over every macrostate $\mathbf{z}$ weighted by its prior belief $p(\mathbf{z})$
- **The computational obstacle**: in $d = 16$ dimensions, numerical quadrature is impossible, and drawing random $\mathbf{z} \sim p(\mathbf{z})$ is a needle in a haystack—almost no prior draw explains a specific $\mathbf{x}$

To evaluate and maximize $\log p_\theta(\mathbf{x})$, we train an encoder $q_\phi(\mathbf{z}|\mathbf{x})$ to find the needle.

A Latent Vector Is Like a Macrostate

Latent-variable model	Statistical mechanics
$\mathbf{z} \sim p(\mathbf{z})$: a few hidden numbers	macrostate, e.g. the magnetisation m
$\mathbf{x} \sim p_\theta(\mathbf{x}\|\mathbf{z})$: a full image, the remaining randomness spread over the pixels	microstate: all N spins, drawn with the bias set by m
$p_\theta(\mathbf{z}\|\mathbf{x})$: which hidden numbers produced this image	measuring the order parameter on a snapshot reads m off $\mathbf{s}$

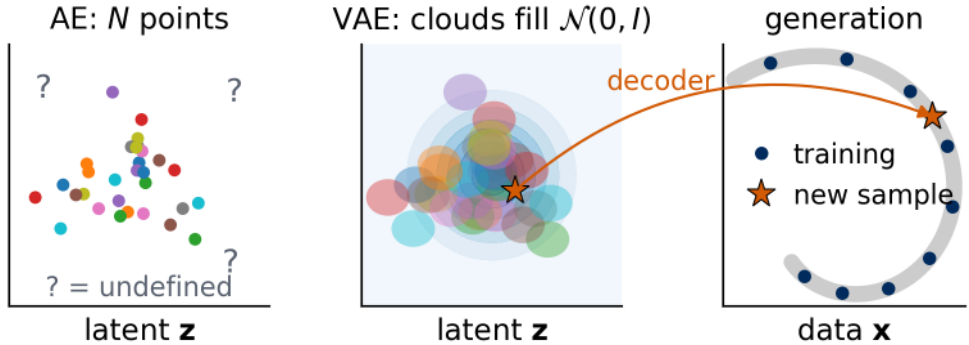
- a few macroscopic numbers fix the character of a configuration; generating = choose them, then fill in the details
- the posterior inverts the generation: it is what the encoder of a VAE approximates
- Lab 025 makes this literal: $\mathbf{x}$ = spin configuration, every spin drawn independently given $\mathbf{z}$ (Section 6)

2

Variational Autoencoders

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From a Discrete Set of Codes to a Continuous Distribution



- AE: N images $\rightarrow$ N latent *points*; between them the decoder is undefined
- wanted: a *continuous* density without holes, to interpolate and to draw new $\mathbf{z} \rightarrow$ new samples
- VAE: a cloud $q_\phi(\mathbf{z}|\mathbf{x})$ per image, all on $\mathcal{N}(0, I)$; new $\mathbf{z}$ (star) $\rightarrow$ decoder $\rightarrow$ new sample

Clouds Instead of Points

The AE failed for two reasons: no distribution on the latent space, and a decoder trained only on isolated codes. The **variational autoencoder** (VAE) changes what is trained:

- **prior** $p(\mathbf{z}) = \mathcal{N}(0, I_d)$, fixed: where latent vectors live
- **decoder** $p_\theta(\mathbf{x}\,|\,\mathbf{z}) = \prod_{j=1}^{784} \hat{x}_j^{x_j}(1-\hat{x}_j)^{1-x_j}$, $\hat{\mathbf{x}} = \sigma(G_\theta(\mathbf{z}))$: independent Bernoulli pixels, $\hat{x}_j$ = probability that pixel j is black
- **encoder** $q_\phi(\mathbf{z}\,|\,\mathbf{x}) = \mathcal{N}(\boldsymbol{\mu}_\phi(\mathbf{x}), \operatorname{diag}\boldsymbol{\sigma}^2_\phi(\mathbf{x}))$: a Gaussian *cloud* of latent vectors per image, approximating the posterior $p_\theta(\mathbf{z}|\mathbf{x})$

Training asks: points of the cloud of $\mathbf{x}$ decode back to $\mathbf{x}$ (decoder learns whole regions), and every cloud stays close to the prior (clouds of all images overlap and fill it).

D. P. Kingma, M. Welling, ICLR 2014, arXiv:1312.6114; D. J. Rezende, S. Mohamed, D. Wierstra, ICML 2014, PMLR 32, 1278

Generation: First z, then x

Generation after training: a sample of $p_\theta(\mathbf{x}) = \int p(\mathbf{z})\, p_\theta(\mathbf{x}|\mathbf{z})\, d\mathbf{z}$, drawn in two steps, first $\mathbf{z}$ from the prior, then $\mathbf{x}$ from the decoder; no encoder.

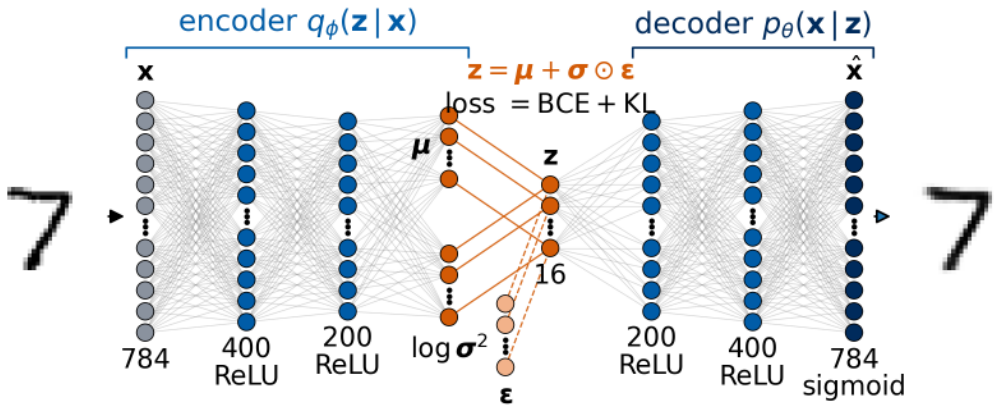
$\mathbf{z} \sim \mathcal{N}(0, I_d)$ → decoder G_θ → $\hat{\mathbf{x}} = \sigma(G_\theta(\mathbf{z}))$ → $x_j \sim \text{Bernoulli}(\hat{x}_j)$, or show $\hat{\mathbf{x}}$

Training needs $\log p_\theta(\mathbf{x})$, an integral over $\mathbb{R}^d$. Two estimates with K samples $\mathbf{z}_k$:

$$\underbrace{\log \frac{1}{K} \sum_{k=1}^{K} p_\theta(\mathbf{x}|\mathbf{z}_k)}_{\mathbf{z}_k \sim \mathcal{N}(0,I)} \qquad \text{or} \qquad \underbrace{\log \frac{1}{K} \sum_{k=1}^{K} \frac{p_\theta(\mathbf{x}|\mathbf{z}_k)\, p(\mathbf{z}_k)}{q_\phi(\mathbf{z}_k|\mathbf{x})}}_{\mathbf{z}_k \sim q_\phi(\mathbf{z}|\mathbf{x}) \text{ (importance sampling)}}$$

$d = 16$, $K = 1000$, mean over 200 test digits: prior -154.8 nats, encoder -93.3 nats. Almost no prior draw lands where $p_\theta(\mathbf{x}|\mathbf{z})$ is large: sampling the prior fails, the encoder is needed.

The VAE Network: Two Heads and a Sampling Node



$\varepsilon \sim \mathcal{N}(0, I)$: a separate input without parameters; $z_j = \mu_j + \sigma_j \varepsilon_j$ element-wise (orange lines).

The Evidence Lower Bound (ELBO)

We cannot compute the evidence $\log p_\theta(\mathbf{x}) = \log \int p(\mathbf{z}) p_\theta(\mathbf{x}|\mathbf{z})\, d\mathbf{z}$. We maximize a **tractable lower bound** instead:

$$\log p_\theta(\mathbf{x}) \;\geq\; \textbf{ELBO} \;=\; \underbrace{\mathbb{E}_{q_\phi(\mathbf{z}|\mathbf{x})}\big[\log p_\theta(\mathbf{x}\,|\,\mathbf{z})\big]}_{\textbf{Reconstruction Accuracy}} \;-\; \underbrace{\mathrm{KL}(q_\phi(\mathbf{z}|\mathbf{x}) \,\|\, p(\mathbf{z}))}_{\textbf{Price of Information}}$$

1. Reconstruction ($-\mathrm{BCE}$)

Codes drawn from the cloud of $\mathbf{x}$ decode back to $\mathbf{x}$.
Forces the decoder to faithfully explain the data.

2. Regularization (KL)

Pulls each cloud towards the prior $\mathcal{N}(0, I)$.
Prevents isolated points; fills the disc so random $\mathbf{z}$ lands on real data.

Maximizing ELBO raises $\log p_\theta(\mathbf{x})$ while pulling the encoder q_ϕ toward the true posterior.

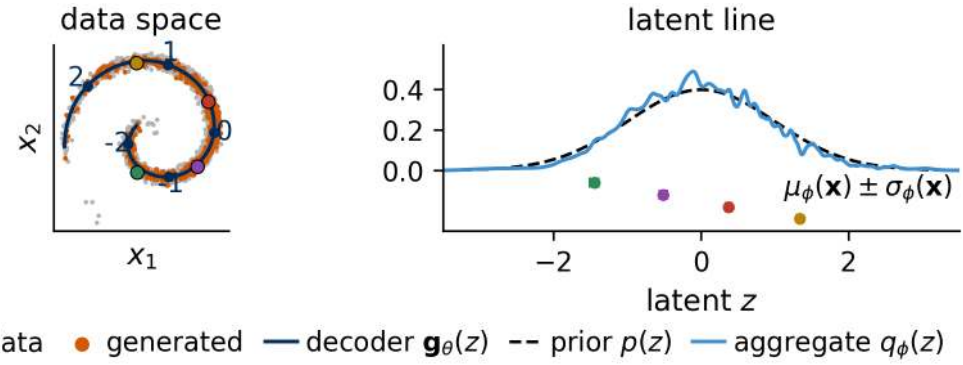
The VAE Loss: Reconstruction and the Price of Information

Loss per image $= -\text{ELBO}$, one sample $\mathbf{z} \sim q_\phi(\mathbf{z}|\mathbf{x})$:

$$\mathcal{L}(\mathbf{x}) = \underbrace{\mathrm{BCE}(\mathbf{x}, \hat{\mathbf{x}})}_{-\log p_\theta(\mathbf{x}|\mathbf{z})} + \underbrace{\frac{1}{2}\sum_{k=1}^{d} (\mu_k^2 + \sigma_k^2 - \ln \sigma_k^2 - 1)}_{\mathrm{KL}(q_\phi(\mathbf{z}|\mathbf{x}) \,\|\, \mathcal{N}(0,I))}, \qquad \hat{\mathbf{x}} = \sigma(G_\theta(\mathbf{z}))$$

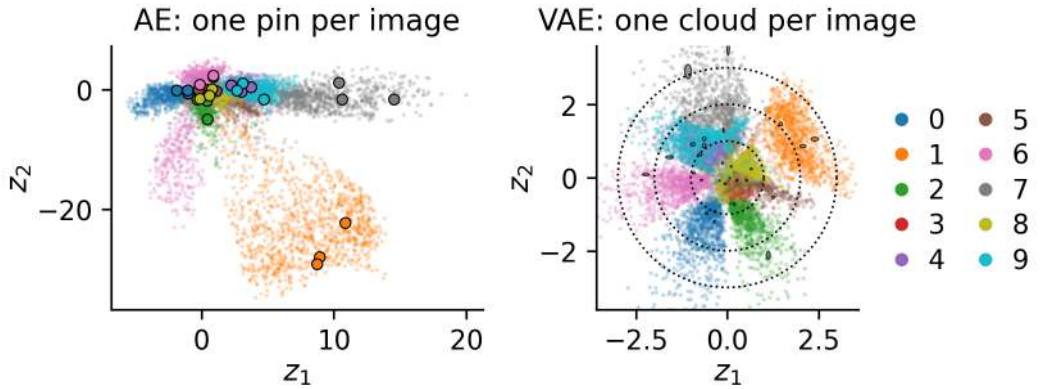
- **binary cross-entropy** $\mathrm{BCE}(\mathbf{x}, \hat{\mathbf{x}}) = -\sum_j [x_j \ln \hat{x}_j + (1 - x_j) \ln(1 - \hat{x}_j)]$: negative log-likelihood of independent Bernoulli pixels with means $\hat{x}_j$; per pixel 0 if certain and correct, $\ln 2 = 0.69$ nats for $\hat{x}_j = \frac{1}{2}$; alone, an autoencoder (points, not clouds)
- **KL term**: zero only for $\boldsymbol{\mu} = 0$, $\boldsymbol{\sigma} = 1$, a cloud that says nothing about $\mathbf{x}$; moving clouds apart (μ_k^2) and narrowing them ($-\ln \sigma_k^2$) costs nats: the price of the information $\mathbf{z}$ carries about $\mathbf{x}$
- clouds pulled onto $\mathcal{N}(0, I)$ cover the prior: a fresh $\mathbf{z} \sim \mathcal{N}(0, I)$ lands where the decoder was trained

A VAE on Points near a Curve



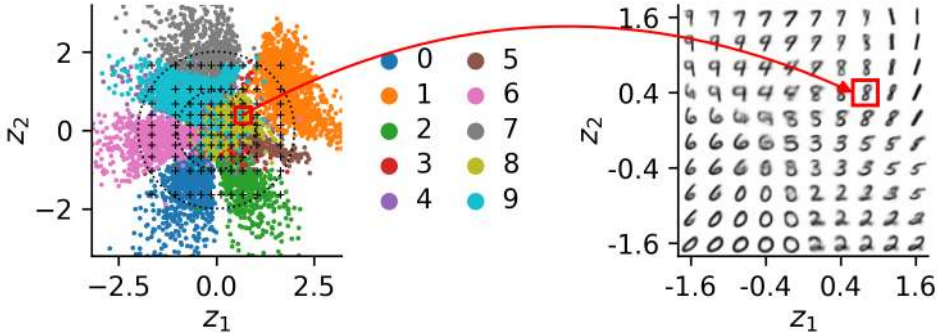
$\mathbf{x} = \mathbf{f}(z_{\text{true}}) + 0.08\,\boldsymbol{\eta}$ on a spiral; VAE with 1D latent, $p_\theta(\mathbf{x}|z) = \mathcal{N}(\mathbf{g}_\theta(z), 0.08^2 I)$. Dots right: clouds $\mu_\phi \pm \sigma_\phi$ of the coloured points (median $\sigma_\phi = 0.04$); **aggregate posterior** $q_\phi(z) = \frac{1}{n}\sum_i q_\phi(z|\mathbf{x}_i) \approx p(z)$; $|\mathrm{corr}(\mu_\phi, z_{\text{true}})| = 0.993$.

The KL Term Turns Islands into One Covered Disc



$d = 2$; dots: all test digits; pins / ellipses $\boldsymbol{\mu}_\phi \pm \boldsymbol{\sigma}_\phi$: three per class; dotted: $|\mathbf{z}| = 1, 2, 3$. Coordinate std: AE 3.8, 7.5, VAE 1.03, 1.09; median $\boldsymbol{\sigma}_\phi = 0.04, 0.04$. AE = a pin per image, islands in an empty sea; VAE = a cloud per image, pulled onto one disc with no sea: **sampling = throwing a dart at the disc**.

Every Grid Point of the VAE Decodes to a Digit



VAE, $d = 2$, layout as for the AE; left: means $\boldsymbol{\mu}_\phi(\mathbf{x})$. **Cell at (z_1, z_2) = decoder mean** $\sigma(G_\theta(\mathbf{z}))$ (grey = pixel probability). Grid: $z^{(j)} = \Phi^{-1}\big((j + \frac{1}{2})/10\big)$, Φ = normal CDF: each cell has prior mass $1/100$.

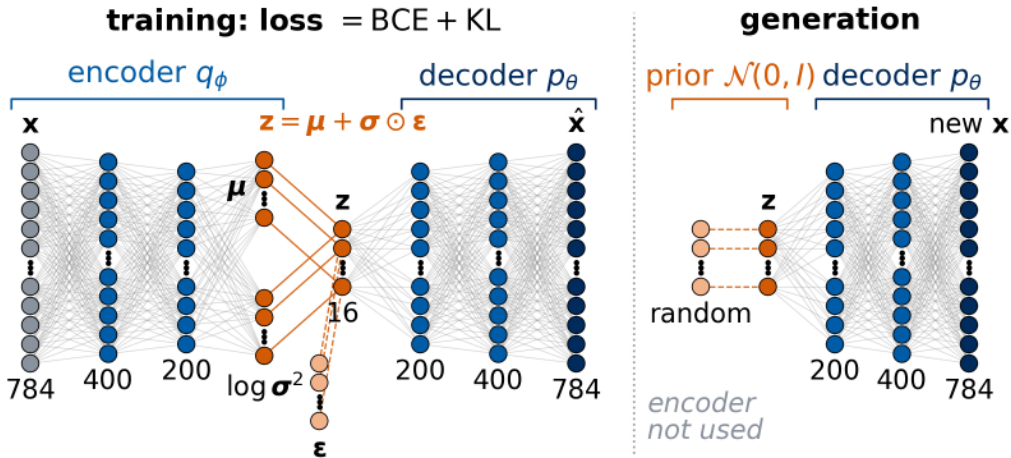
- clouds cover the prior disc: no nonsense cells; one region per digit, morphing at borders
- blurry: the mean averages all pixel patterns compatible with $\mathbf{z}$

3

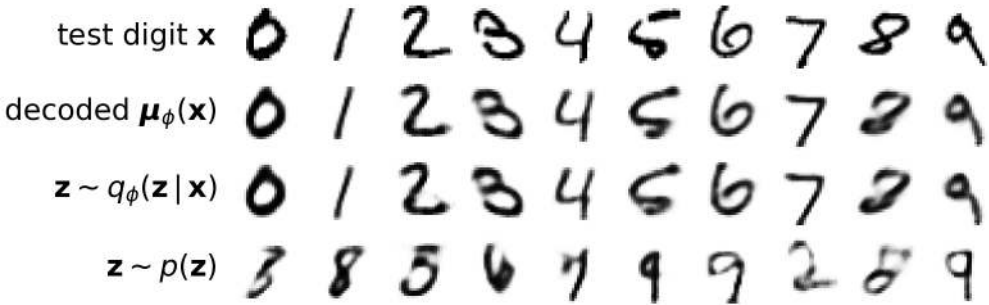
Sampling a Trained VAE

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Training and Generation Use Different Parts of the VAE

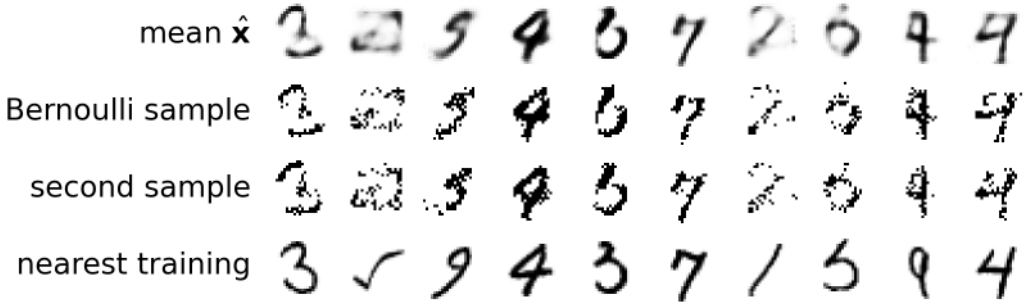


Reconstruction and Generation



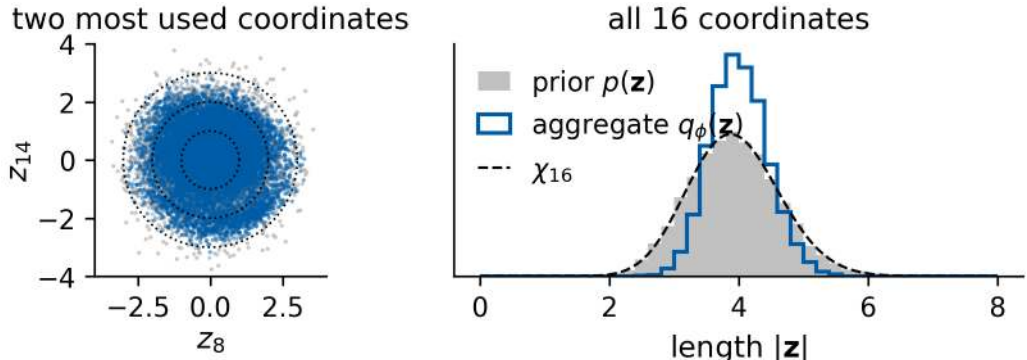
- **reconstruction** $\mathbf{x} \to \mathbf{z} \sim q_\phi(\mathbf{z}|\mathbf{x}) \to p_\theta(\mathbf{x}|\mathbf{z})$: identity and most of the style kept
- **generation** $\mathbf{z} \sim p(\mathbf{z}) \to p_\theta(\mathbf{x}|\mathbf{z})$: only decoder and prior are kept
- $d = 16$, decoder means; confidence > 0.9 of a classifier (Section 4): real 96%, reconstr. 92%, generated 54%

A Sample Is Not the Decoder Mean



- shown so far: the mean $\hat{x}_j$ (blurry); a **sample** draws each pixel (black with prob. $\hat{x}_j$): speckled
- correlations (a continuous stroke) pass through $\mathbf{z}$; the rest are coin flips, as for spins (Section 6)
- not copies: distance to the nearest training image 4.48 (generated), 4.19 (unseen test)

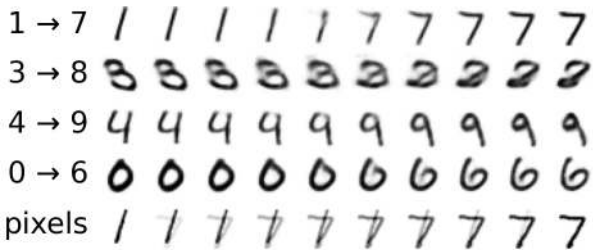
The Encoded Data Do Not Fill the Prior Exactly



Aggregate posterior $q_\phi(\mathbf{z}) = \frac{1}{n}\sum_i q_\phi(\mathbf{z}|\mathbf{x}_i)$: where the encoded data live (sampled: encode a test image, draw from its cloud). Grey: $\mathcal{N}(0, I)$; χ_{16}: distribution of $|\mathbf{z}|$ under the prior.

- per-coordinate std of $q_\phi(\mathbf{z})$ between 0.97 and 1.04; mean $|\mathbf{z}|$: 4.01 (q_ϕ), 3.93 (prior)
- median distance to the nearest test code: prior samples 2.61, other codes 1.55: **prior holes**

Interpolation in the Latent Space

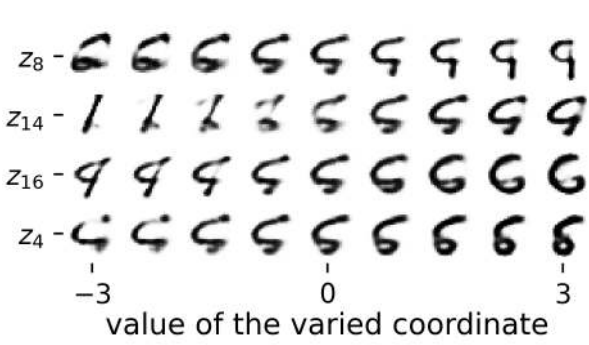


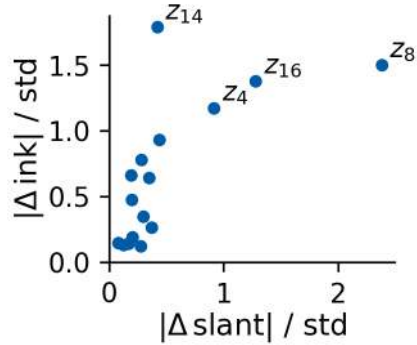
$\mathbf{z}(t) = (1-t)\,\boldsymbol{\mu}_\phi(\mathbf{x}_a) + t\,\boldsymbol{\mu}_\phi(\mathbf{x}_b)$, $t \in [0,1]$, decoded by the $d = 16$ VAE; last row: $(1-t)\,\mathbf{x}_a + t\,\mathbf{x}_b$.

- latent line: plausible digits throughout; the 1 bends into a 7, the 4 closes into a 9
- pixel line: two superposed digits

Straight lines in the latent space stay among the data.

One Latent Coordinate Changes Several Properties at Once





Left: $\boldsymbol{\mu}_\phi$ of one test 5, one coordinate varied from -3 to 3 (four most used, by $\overline{\mathrm{KL}_k} = \langle \mathrm{KL} \text{ of coordinate } k \rangle_{\mathbf{x}}$, from 0.65 to 2.23 nats). Right: change of slant $-\mathrm{Cov}(u, v)/\mathrm{Var}(v)$ and of ink (mean pixel value) for $z_k = -2 \to 2$, 300 test digits, in units of their spread over real digits; one dot per coordinate. 7 of 16 coordinates change both by > 0.3.

4

Judging Synthetic Data



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A Classifier as a Referee

Pictures are judged by eye. To *measure* what a generator produces, use a referee trained on the **real** data, with known accuracy on real test data, and apply it unchanged to synthetic images.

- CNN: 3×3 convolutions with 16 and 32 channels, ReLU, each followed by 2×2 max-pooling ($28 \to 14 \to 7$); dense 128; 10 outputs $f_c(\mathbf{x})$
- trained with the cross-entropy, the negative log-likelihood of the labels (Adam, 4 epochs):

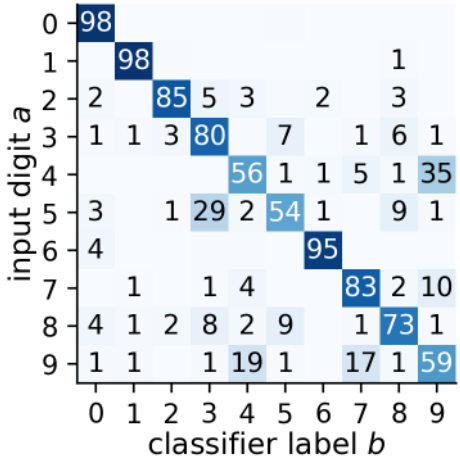
$$\mathcal{L} = -\frac{1}{n} \sum_i \log p(y_i \,|\, \mathbf{x}_i), \qquad p(c \,|\, \mathbf{x}) = \frac{e^{f_c(\mathbf{x})}}{\sum_{c'} e^{f_{c'}(\mathbf{x})}}$$

test accuracy on 10 000 real digits

0.9854

Every test of synthetic data is first run on real data: the baseline.

Confusion Matrix of VAE Reconstructions



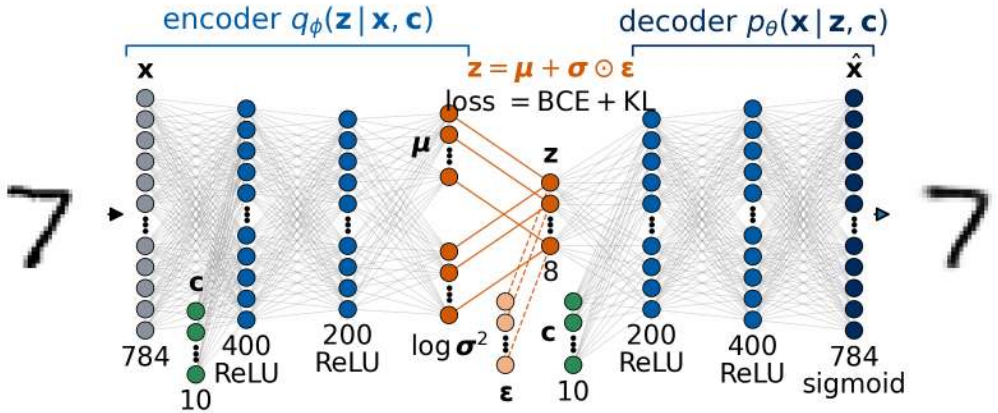
Row a: digit of the real test input $\mathbf{x}$; column b: classifier label of the reconstruction $\sigma(G_\theta(\mathbf{z}))$, $\mathbf{z} \sim q_\phi(\mathbf{z} \mid \mathbf{x})$; C_{ab} in % of row a. Shown: $d = 2$.

- mean diagonal $\frac{1}{10}\mathrm{tr}\, C$: real images 0.985, $d = 16$ 0.953, $d = 2$ 0.783
- $d = 16$: the digit survives the latent space
- $d = 2$: look-alike digits share a region of the plane; worst: $4 \to 9$: 35%

5
Conditional Generation

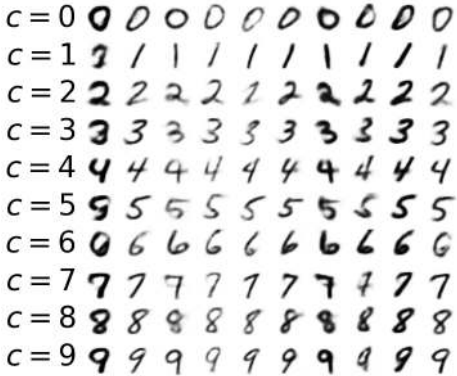
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The Conditional VAE Receives the Label as an Input



Known, labelled factor $\Rightarrow$ give it to both networks: conditional VAE (cVAE); $\mathbf{c}$ = one-hot digit, $c_j = \delta_{j,y}$ (a single number would put 3 "between" 2 and 4); $d = 8$.

Digits on Demand

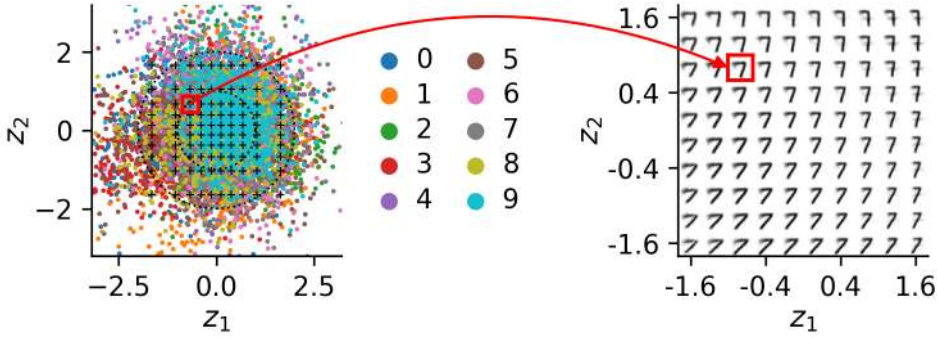


$\sigma(G_\theta(\mathbf{z}, \mathbf{c}))$; row: $\mathbf{c}$ = digit 0 to 9; column: one $\mathbf{z} \sim \mathcal{N}(0, I)$, the same in every row. Along a column the digit changes and the style stays.



Style transfer: a real test digit $\mathbf{x}$ (left), $\mathbf{z} = \boldsymbol{\mu}_\phi(\mathbf{x}, \mathbf{c}_y)$, decoded with $\mathbf{c} = 0, \ldots, 9$.

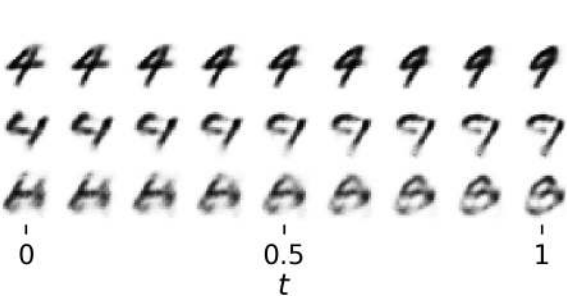
The Latent Plane of a cVAE Holds Only the Style

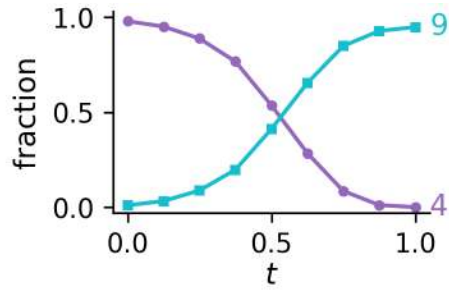


cVAE, $d = 2$, layout and quantile grid as before. Left: $\boldsymbol{\mu}_\phi(\mathbf{x}, \mathbf{c})$ of the test digits, coloured by label; right: **cell = decoder mean with $\mathbf{c} = 7$** at that grid point.

- all digits share one disc: digit predicted from the nearest class mean in the latent plane, VAE 0.73, cVAE 0.18 (chance 0.1)
- every cell is a 7; moving across the grid changes only how it is written

A Condition Never Seen in Training Is Interpolated

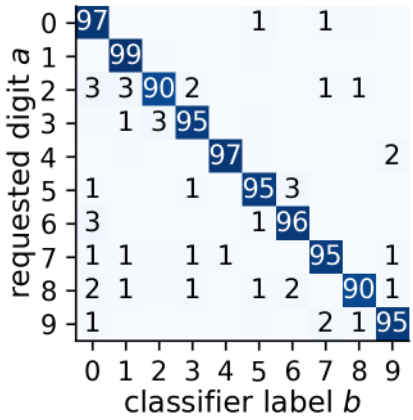


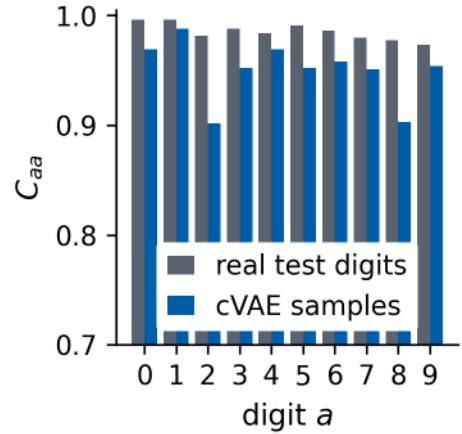


$\mathbf{c}(t) = (1-t)\,\mathbf{e}_4 + t\,\mathbf{e}_9$ ($\mathbf{e}_k$: one-hot vector of digit k); left: three fixed $\mathbf{z}$, right: fraction of 1000 samples, $\mathbf{z} \sim \mathcal{N}(0, I)$, that the classifier assigns to 4 and to 9.

- only $t = 0, 1$ were seen; in between the decoder interpolates, the style of each row is kept
- for digits there is no "6.5"; for a temperature or a point (g, T) intermediate values exist, and interpolated samples can be checked against Monte Carlo (Section 6)

The Requested Digit Is Generated





1000 samples per requested digit, $\mathbf{z} \sim \mathcal{N}(0, I)$;
$C_{ab} = \#\{\text{samples generated with } \mathbf{c} = a \text{ assigned to } b\}/1000$, in %. Right: C_{aa} next to the classifier on real test digits. Mean C_{aa}: 0.950; largest confusion $2 \to 0$: 3%.

From Digits to Spins

The protocol of this part, ready for physics:

1. a generator conditioned on what we want: $\mathbf{c}$ = the digit $\longrightarrow$ $\mathbf{c} = T$ for the 2D Ising model, $\mathbf{c} = (g, T)$ for the J_1–J_2 model
2. referees trained on **real** data, checked on real test data first $\longrightarrow$ networks trained on Monte Carlo configurations
3. the same tests on synthetic samples

Physics has a stricter referee

the true distribution is known, $p(\mathbf{s} \mid T) = e^{-H(\mathbf{s})/T}/Z$: energy, magnetisation, fluctuations and correlations of generated configurations can be compared with Monte Carlo at every T

The test

cVAE samples against the Monte Carlo samples they were trained on, observable by observable, including near the phase transition

6

Generated Spins and the Hamiltonian

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The Hamiltonian Fixes the Distribution

spins $s_i = \pm 1$, $L \times L$ lattice, periodic boundaries, $N = L^2$, $k_B = J = J_1 = 1$, $\beta = 1/T$:

$$H_{\text{Ising}} = -\sum_{\langle ij \rangle} s_i s_j, \quad H_{J_1 J_2} = -\sum_{\langle ij \rangle} s_i s_j + g \sum_{\langle\langle ij \rangle\rangle} s_i s_j, \quad p(\mathbf{s} \mid T) = \frac{e^{-\beta H(\mathbf{s})}}{Z}$$



J_1, $J_2 = g$

ground states ($T = 0$), $2N$ nearest and $2N$ diagonal bonds: ferromagnet $e = -2 + 2g$; stripes (sketch: nearest bonds cancel, diagonals antiparallel) $e = -2g$; equal at $g = \frac{1}{2}$: ferromagnet for $g < \frac{1}{2}$, stripes for $g > \frac{1}{2}$

2D Ising ($g = 0$): $T_c = 2/\ln(1 + \sqrt{2}) = 2.269$ (L. Onsager, Phys. Rev. **65**, 117 (1944); C. N. Yang, Phys. Rev. **85**, 808 (1952)). A generator of spin configurations has to reproduce $p(\mathbf{s} \mid T)$, not only its look.

Observables: Definition and Estimator from Samples

Physical observables are averages over the Boltzmann distribution $p(\mathbf{s}|T) = e^{-\beta H(\mathbf{s})}/Z$. Summing 2^N states is intractable $\Longrightarrow$ Monte Carlo draws n snapshots $\mathbf{s}^{(k)} \sim p$, $\langle O \rangle \approx \overline{O} = \frac{1}{n}\sum_k O(\mathbf{s}^{(k)})$. A generative model is judged by the **same sample estimators**.

	definition through Z	estimator from n samples	measures
energy	$\langle e \rangle = -\frac{1}{N}\partial_\beta \ln Z$	$\overline{e} = \frac{1}{n}\sum_k H(\mathbf{s}^{(k)})/N$	bonds
magnetisation	$\langle \lvert m \rvert \rangle = \frac{1}{Z}\sum_{\mathbf{s}} \lvert m \rvert\, e^{-\beta H}$	$\overline{\lvert m \rvert} = \frac{1}{n}\sum_k \lvert m_k \rvert$	ferromagnetism
stripe order	$\langle m_{\mathrm{str}} \rangle = \left(m^2_{(\pi,0)} + m^2_{(0,\pi)}\right)^{1/2}$	$\overline{m_{\mathrm{str}}}$	stripes
specific heat	$c = N\beta^2\left(\langle e^2 \rangle - \langle e \rangle^2\right)$	$N\beta^2\left(\overline{e^2} - \overline{e}^2\right)$	e fluct.
susceptibility	$\chi = N\beta\left(\langle m^2 \rangle - \langle \lvert m \rvert \rangle^2\right)$	$N\beta\left(\overline{m^2} - \overline{\lvert m \rvert}^2\right)$	m fluct.
Binder cumulant	$U_4 = 1 - \langle m^4 \rangle / 3\langle m^2 \rangle^2$	$1 - \overline{m^4}/3\,\overline{m^2}^2$	$P(m)$ shape
correlation	$G(r) = \frac{1}{2N}\sum_{\mathbf{x},\mu}\langle s_{\mathbf{x}} s_{\mathbf{x}+r\hat{\mathbf{e}}_\mu} \rangle$	same with $\overline{\cdot}$	range
distributions	$P(e') = \frac{1}{Z}\sum_{\mathbf{s}} \delta\left(e' - H/N\right) e^{-\beta H}$	histogram of e_k (and m_k)	all moments

The cVAE for Spins: a Product of Bernoulli Factors

condition $c = T$ (Ising) or $c = (g, T)$ (J_1–J_2); bits $b_i = (s_i + 1)/2$; one logit $\ell_i(\mathbf{z}, c)$ per site, $\sigma(\ell_i)$ = probability that spin i points up

$$\log p_\theta(\mathbf{s} \mid c) \;\geq\; \mathbb{E}_{q_\phi(\mathbf{z}|\mathbf{s},c)}\Big[\sum_{i=1}^{N} b_i \log \sigma(\ell_i) + (1 - b_i) \log\Big(1 - \sigma(\ell_i)\Big)\Big]$$
$$- \,\mathrm{KL}\Big(q_\phi(\mathbf{z} \mid \mathbf{s}, c)\,\Big\|\, p(\mathbf{z} \mid c)\Big) \;=\; \mathrm{ELBO}$$

the first term is $-$BCE of the spins ($N \ln 2 = 624$ nats for coin flips, $N = 900$); prior $p(\mathbf{z}|c) = \mathcal{N}(0, I)$ as in the MNIST cVAE, or learned (below)

trained Ising models ($L = 30$, $d = 16$): $-$ELBO $\approx$ 363–391 nats per configuration (training, held-out), against $N \ln 2 = 624$ for independent coin flips

D. P. Kingma, M. Welling, arXiv:1312.6114; K. Sohn, H. Lee, X. Yan, NeurIPS 2015, NeurIPS proceedings

The cVAE and Monte Carlo Sample Spins Differently

Generating at temperature T

$\mathbf{z} \sim p(\mathbf{z} \mid T)$, logits $\ell_i(\mathbf{z}, T)$, then every spin *independently*: $s_i = +1$ with probability $\sigma(\ell_i)$ (Bernoulli), or $\hat{s}_i = \operatorname{sign} \ell_i$ (most likely). Given $\mathbf{z}$ a mean-field-like product, $\langle s_i \rangle_{\mathbf{z}} = \tanh(\ell_i/2)$,

$$\operatorname{Cov}(s_i, s_j) = \operatorname{Cov}_{\mathbf{z}}\left(\tanh \tfrac{\ell_i}{2}, \tanh \tfrac{\ell_j}{2}\right), \; i \neq j$$

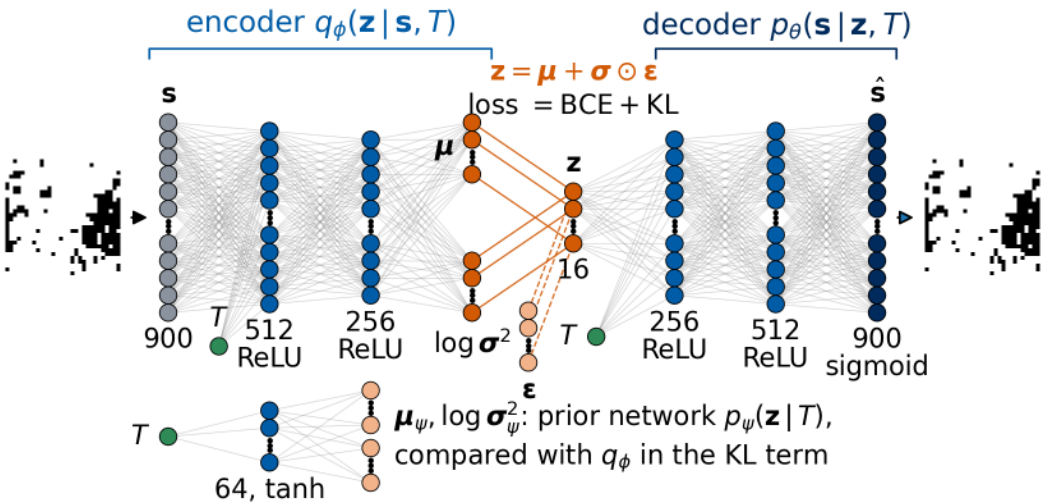
all correlations pass through the $d = 16$ latent variables

R. H. Swendsen, J.-S. Wang, PRL **58**, 86 (1987)

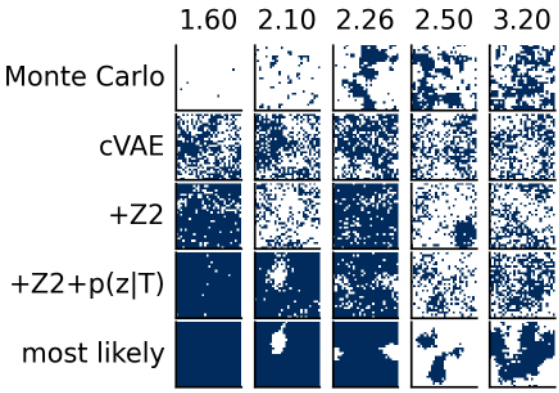
Monte Carlo

a Markov chain of updates built from H (Metropolis single-spin flips, accepted with $\min(1, e^{-\beta \Delta H})$; Ising data here: Swendsen–Wang cluster flips); stationary distribution exactly $e^{-\beta H}/Z$, with every neighbour correlation H implies; costs equilibration, successive snapshots correlated. $H = -\sum s_i s_j$: independent draws break bonds at random $\Rightarrow$ e, c, $G(1)$ are the sharp tests

The Spin cVAE with a Prior Network



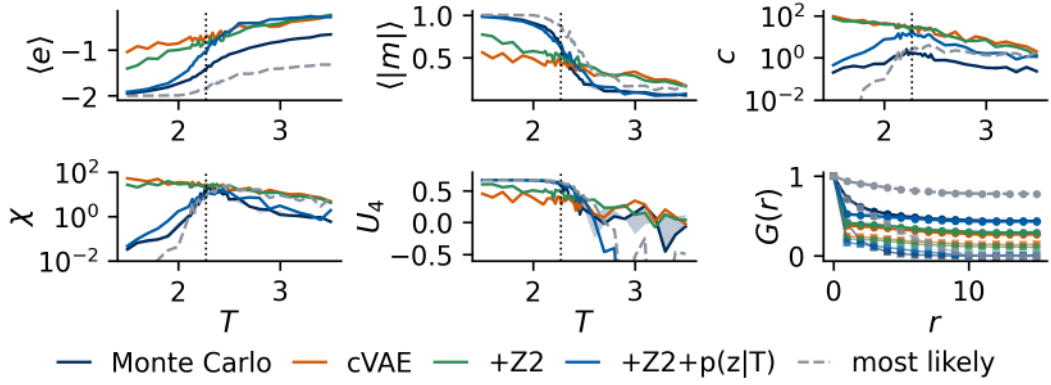
Monte Carlo and Synthetic Ising Configurations



rows: Monte Carlo; cVAE; cVAE+Z2; cVAE+Z2+p(z|T) with Bernoulli sampling; the same $\mathbf{z}$ with $\hat{s}_i = \operatorname{sign} \ell_i$

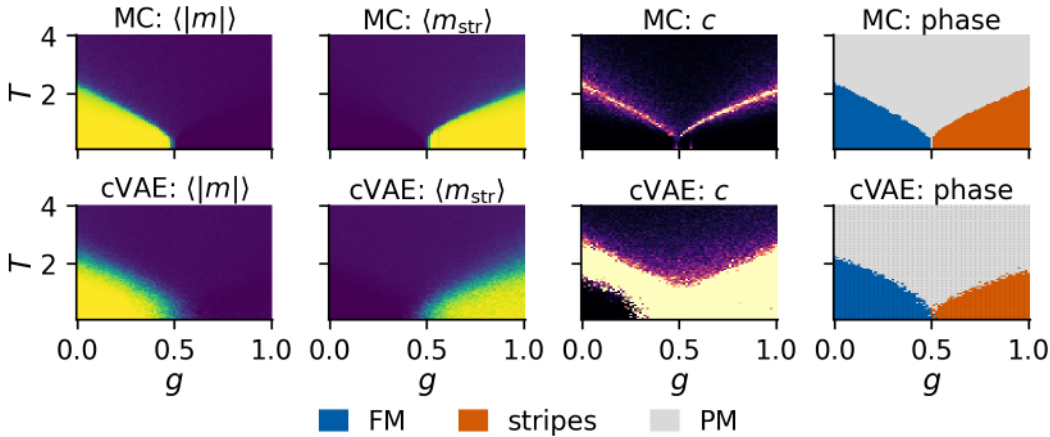
- cVAE and +Z2: the requested T is barely visible
- +p(z|T): ordered below T_c, disordered above; salt-and-pepper noise inside domains
- most likely configuration: no noise, but large smooth domains even at $T = 3.2$

The Order Parameter Is Reproduced, the Energy Is Not



$\langle e \rangle$, $\langle |m| \rangle$, c, χ, U_4 vs T; $G(r)$ at $T = 2.26$ (circles) and $T = 3.0$ (squares); band: Monte Carlo A vs B. Noise ratios of +p(z|T): $R_{|m|} = 2.8$, $R_\chi = 2.0$, $R_{W_1(m)} = 1.3$; $R_e = 34.8$, $R_c = 39.0$, $R_{W_1(e)} = 21.6$. At $T = 2.26$: $\langle e \rangle = -1.43$ (Monte Carlo), -1.04 (Bernoulli), -1.86 (most likely); $G(1) = -\langle e \rangle / 2$: the energy is the nearest-neighbour correlation

J_1–J_2: the Phase Diagram Is Interpolated



Trained at every second g, T (dots); colour 0 to 1 (c: to 2.2); phase: CNN vote. Same phase as MC A: MC B 99.7 %, cVAE 95.3 % (seen) / 95.4 % (unseen), most likely 35.4 %.

The Generator Reproduces the Phase

Reproduced within a few noise units

- Ising $\langle|m|\rangle$, χ, $P(m)$: $R = 2.8, 2.0, 1.3$; J_1–J_2 $\langle|m|\rangle$, $\langle m_{\mathrm{str}}\rangle$: $R \approx 4.8, 5.9$ (unseen)
- the temperature seen by a regressor below T_c, the phase seen by classifiers trained on real data
- the J_1–J_2 phase diagram at unseen (g, T), both stripe orientations; new configurations, not copies

needed: $\mathbb{Z}_2$ symmetry of H by construction, and a conditional prior $p(\mathbf{z}|c)$. Ising; mixed: fraction of snapshots at $T < T_c$ with $|m| < \langle|m|\rangle_{\mathrm{MC}}/2$; Monte Carlo: RMSE 0.12, mixed 1 %

	$R_{\lvert m\rvert}$	R_χ	$R_{W_1(m)}$	RMSE $\hat{T}$	mixed
cVAE	13.3	11.8	3.3	0.98	45 %
cVAE+Z2	10.8	9.7	2.8	0.84	27 %
+p(z|T)	2.8	2.0	1.3	0.62	1 %

The Generator Does Not Reproduce the Energy

Not reproduced

- $\langle e \rangle = H/N$: $R = 34.8$ (Ising), 63.9 (J_1–J_2)
- c: 2 to 15 times the Monte Carlo value (Ising, six tabulated T); $P(e)$ too broad
- a C2ST separates real from synthetic at the three tested T (accuracy 0.73–0.98)

given $\mathbf{z}$ the decoder is a product over sites, a mean-field-like distribution: nearest-neighbour correlations beyond what 16 latent variables carry are lost. Bernoulli sampling breaks bonds, $\operatorname{sign} \ell_i$ over-smooths.

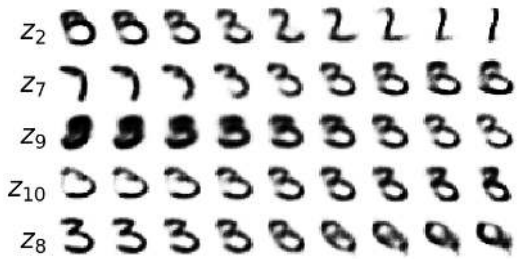
Looking right is not obeying H: synthetic configurations answer questions about the phase; they are not samples of $e^{-H/T}/Z$ and do not replace Monte Carlo for thermodynamics.

7

Disentangled Representations: the β-VAE

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In a Plain VAE One Coordinate Changes Several Features



VAE, $d = 10$: $\boldsymbol{\mu}_\phi(\mathbf{x})$ of one test 3; row: one z_k from -3 to 3, others fixed

Latent traversal: sampling along one axis, $\mathbf{z} = (\mu_1, \ldots, \mu_{k-1}, t, \mu_{k+1}, \ldots)$, $t \in [-3, 3]$, decoded.

- z_2: $3 \to 2 \to 1$ with slant; z_7: $7 \to 3 \to 8$; only z_9 changes one property (thickness)
- tells: what the decoder does along this axis *near this code*
- does not tell: that the axis means the same everywhere, or that no other axis changes the same feature

The Goal: One Coordinate per Independent Factor

Generative factors: the few independent quantities that produced an image.
sprite: shape, scale, rotation, x, y · face: pose (looking left or right), glasses, smile, hair · digit: identity, slant, thickness · spins: magnetisation m, stripe orientation

Disentangled code $\mathbf{z} = (z_1, \ldots, z_d)$: each factor is carried by its own coordinate; moving one z_k changes one property of the image, and changing one factor changes one z_k.

- *control* (add glasses by moving one number), *interpretation*, *discovery* (an order parameter without labels)
- cVAE: the factor is *given* as a label $\mathbf{c}$; here the network must *discover* it

MNIST	watch the loss at work: rate, distortion, samples, coordinates switching off
dSprites	synthetic sprites with 5 known factors: *measure* disentangling
CelebA	CelebFaces Attributes: celebrity faces with 40 labels; traversals, arithmetic
spins	Ising, J_1–J_2: which coordinate is the magnetisation, the energy, the stripes?

The β-VAE Loss, Term by Term

$$\mathcal{L}_\beta(\mathbf{x}) = \underbrace{\mathbb{E}_{q_\phi(\mathbf{z}|\mathbf{x})}\left[\mathrm{BCE}(\mathbf{x}, \hat{\mathbf{x}})\right]}_{\text{reconstruction error}} + \beta \underbrace{\mathrm{KL}\left(q_\phi(\mathbf{z}|\mathbf{x}) \,\|\, p(\mathbf{z})\right)}_{\text{code vs. prior}}, \qquad \beta = 1: \;\; \mathcal{L}_1 = -\mathrm{ELBO}$$

- reconstruction: samples of the cloud of $\mathbf{x}$ decode back to $\mathbf{x}$; KL: each cloud pulled to $\mathcal{N}(0, 1)$
- $\beta > 1$: information costs more $\Rightarrow$ fewer used coordinates, wider clouds, blurrier images, stronger pull to the *factorised* prior $\prod_k \mathcal{N}(0, 1)$: towards independent coordinates
- $\beta < 1$: sharper, but the code drifts from the prior; $\beta \to 0$: an autoencoder
- $\beta =$ Lagrange multiplier: best reconstruction at $\mathrm{KL} < \varepsilon$ (not $1/T$)

I. Higgins et al., ICLR 2017, OpenReview

Rate and Distortion: β Is Minus the Slope

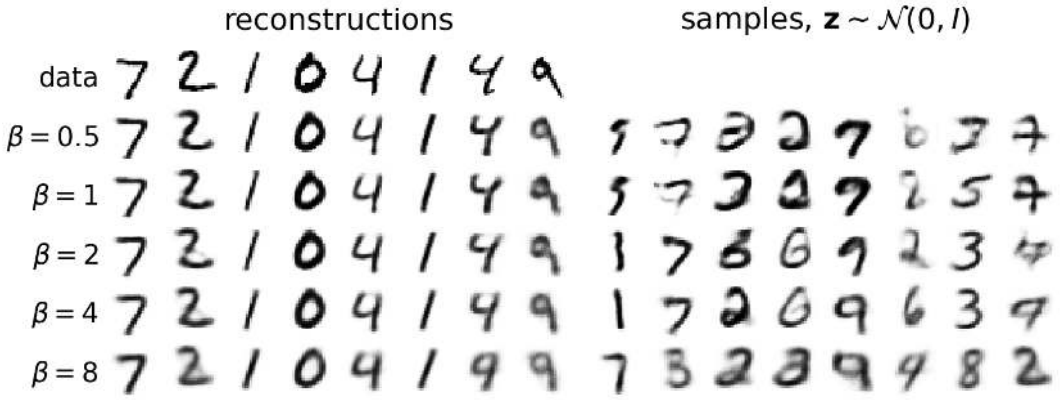
Averages over the data $p_{\mathrm{data}}(\mathbf{x})$, in nats per image:

$$D = \mathbb{E}_{p_{\mathrm{data}}}\mathbb{E}_{q_\phi}\big[-\log p_\theta(\mathbf{x}|\mathbf{z})\big], \qquad R = \mathbb{E}_{p_{\mathrm{data}}}\mathrm{KL}(q_\phi(\mathbf{z}|\mathbf{x}) \,\|\, p(\mathbf{z})), \qquad \mathcal{L}_\beta = D + \beta R$$

- **rate** R: the information the code carries about the image; **distortion** D: the reconstruction error
- every trained model is a point (R, D); the reachable points lie above a decreasing curve $D^*(R)$
- minimising $D + \beta R$ selects the point where $dD^*/dR = -\beta$: β is minus the slope
- large β: the flat, low-rate end; small β: the steep, high-rate end
- the ELBO $= -(D + R)$ alone does not fix how a model splits its loss between R and D

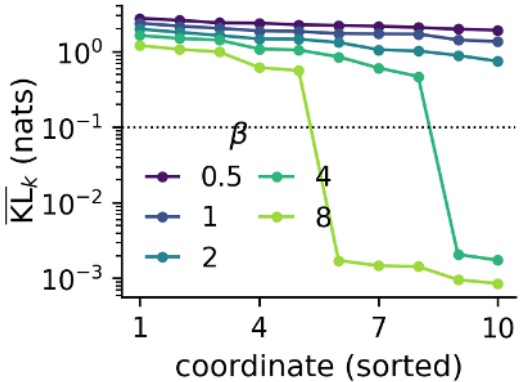
A. A. Alemi et al., *Fixing a Broken ELBO*, ICML 2018, PMLR 80, 159

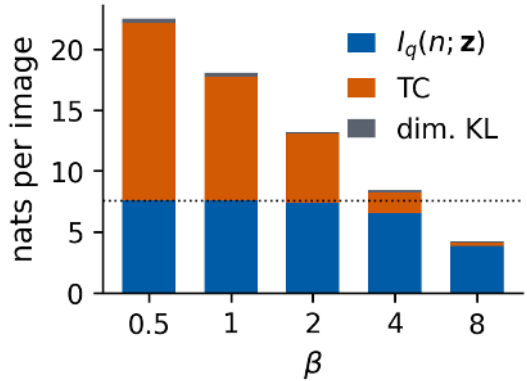
Larger β Blurs Reconstructions and Samples



Left: test digits and $\sigma(G_\theta(\boldsymbol{\mu}_\phi(\mathbf{x})))$; right: $\sigma(G_\theta(\mathbf{z}))$, $\mathbf{z} \sim \mathcal{N}(0, I)$, the *same* 8 vectors $\mathbf{z}$ for every β. $\beta = 8$: detail lost (the 4 becomes a 9); the same $\mathbf{z}$ decodes to different digits in different models; $\beta = 0.5, 1$: some malformed samples.

Larger β Switches Coordinates Off and Removes Total Correlation



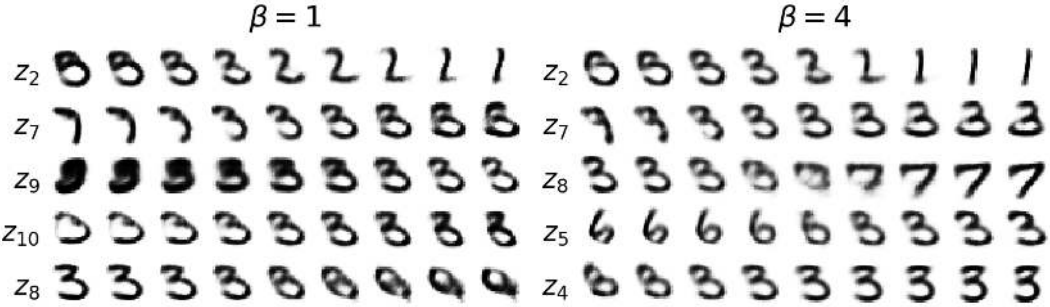


$\overline{\mathrm{KL}_k}$ sorted; active above 0.1 nats

parts of R on $M = 2000$ digits; dotted: $\log M$

Active: 10, 10, 10, 8, 5 of 10; TC: 14.50 $\to$ 0.26 nats; $I_q \approx \log M$ up to $\beta = 2$; parts add up to the closed-form R within 0.01 nats.

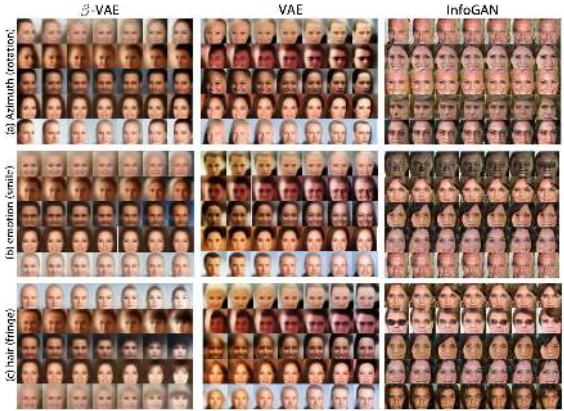
On MNIST, Traversals Show Style, Not Digit Identity



Same test 3; row: one active z_k from -3 to 3, others fixed.

- both models change the identity along every axis ($\beta = 4$: z_2: $6 \to 3 \to 1$, z_8: $3 \to 7$); $\beta = 4$ has fewer, somewhat cleaner axes (z_4: width, thickness)
- identity is *discrete*, $\mathbf{z}$ continuous: digits morph (cVAE: identity as a label); no true factors listed: disentangling cannot be *measured* on MNIST

A β-VAE Turns Heads Where a Plain VAE Mixes Factors

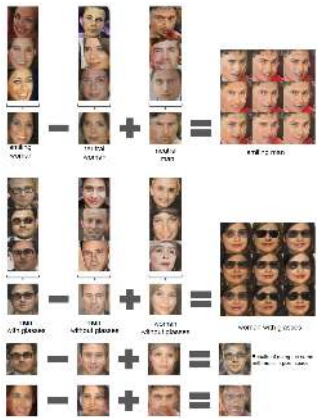


CelebA at 64×64; block: one coordinate from -3 to 3; row: one seed face

- β-VAE ($\beta = 250$): azimuth (left/right), smile, fringe, each in one coordinate
- VAE ($\beta = 1$): azimuth mixed with emotion, glasses, gender
- InfoGAN: a GAN-based method
- sunglasses were not isolated

Fig. 1 of I. Higgins et al., *β-VAE: Learning Basic Visual Concepts with a Constrained Variational Framework*, ICLR 2017, OpenReview; CelebA: Z. Liu, P. Luo, X. Wang, X. Tang, ICCV 2015, doi:10.1109/ICCV.2015.425

Latent Arithmetic: "Man with Glasses − Man + Woman"



- column: latent vectors of 3 faces, averaged
- $\bar{\mathbf{z}}_{\text{man, glasses}} - \bar{\mathbf{z}}_{\text{man}} + \bar{\mathbf{z}}_{\text{woman}}$, decoded: a woman with glasses
- bottom rows: the same on pixels, misaligned overlaps
- a **GAN** (generative adversarial network), not a VAE, on web faces, not CelebA

Figure 7 of A. Radford, L. Metz, S. Chintala, *Unsupervised Representation Learning with Deep Convolutional Generative Adversarial Networks*, arXiv:1511.06434

A Latent Vector Rotates Faces from Left to Right



Latent difference between faces looking left and faces looking right (four of each, averaged), added in steps to the latent vectors of random generated faces (rows): the pose changes, the person largely stays. The generator is the deep convolutional GAN of Radford, Metz and Chintala. Figure 8 of A. Radford, L. Metz, S. Chintala, arXiv:1511.06434

Reconstructions Keep the Global Appearance, Not the Details



top: test faces $\mathbf{x}$; bottom: $\sigma(G_\theta(\boldsymbol{\mu}_\phi(\mathbf{x})))$

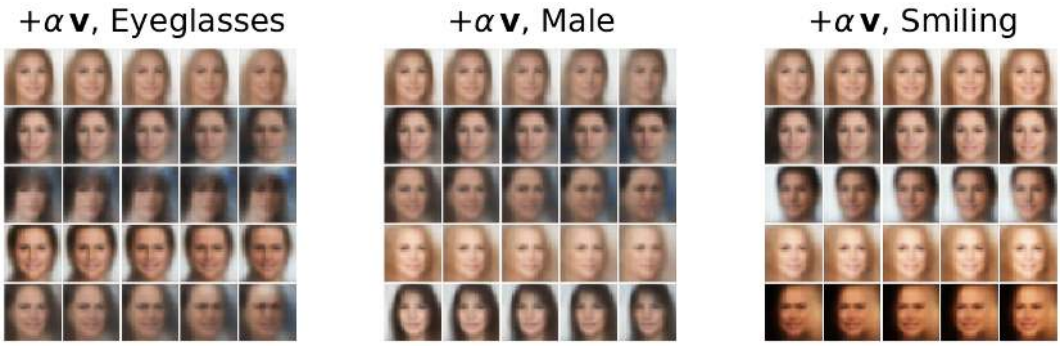
- test rate $R = 15.8$ nats per face; 18 of 32 coordinates active
- kept: pose, skin and hair colour, background, shape of the face
- lost: eyes, mouth, glasses; 15.8 nats is a small budget for a 32×32 colour image, blur is the price

Our Faces Model Separates Light, Skin and Hair, Not Pose



Two test faces (left, right block); row: one z_k from -3 to 3, 8 largest $\overline{\mathrm{KL}}_k$. Label: attribute a with the largest $|\rho(\mu_k, a)|$ over the training faces, ρ = Pearson correlation of μ_k with the 0/1 label. z_{10}: background; z_{32}: direction of the light; z_{25}: skin and hair colour; z_{23}: short-haired man $\rightarrow$ long-haired woman; z_2: hair length. No coordinate turns the head cleanly; $|\rho| \leq 0.35$: the labels describe details, the coordinates global appearance.

Attribute Directions Exist but Are Faint at 32×32



$\mathbf{v}_a = \langle \boldsymbol{\mu}_\phi(\mathbf{x}) \rangle_{a=1} - \langle \boldsymbol{\mu}_\phi(\mathbf{x}) \rangle_{a=0}$ (labels used only after training); rows: test faces without a, decoded $\boldsymbol{\mu}_\phi(\mathbf{x}) + \alpha\, \mathbf{v}_a$, $\alpha = 0, 0.5, 1, 1.5, 2$. Glasses: a dark band at the eyes for $\alpha = 2$; male: darker, broader face, shorter hair; smiling: brighter lower face.

8

Physics Learned by the β-VAE

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From Generating to Understanding

So far a VAE was judged by what it draws. Now we use it to learn about the *system*: the network sees only snapshots $\mathbf{s}$, no T, no phase, no Hamiltonian, and compresses each into a few numbers that β pushes to be independent. If the compression is good, these numbers are the quantities in which snapshots differ most: the macroscopic ones.

- class activation maps showed *where* a classifier looks
- the first principal component of Ising snapshots is the magnetisation (Wang 2016)
- a VAE latent variable correlates with the magnetisation (Wetzel 2017)
- a β-VAE cost function recovers spring constant and damping of a pendulum from its trajectory (Iten et al. 2020)

Known model: a check. Unknown model: a discovery tool. L. Wang, Phys. Rev. B **94**, 195105 (2016), doi; S. J. Wetzel, Phys. Rev. E **96**, 022140 (2017), doi; R. Iten, T. Metger, H. Wilming, L. del Rio, R. Renner, Phys. Rev. Lett. **124**, 010508 (2020), doi

The Probing Protocol

1. train a β-VAE on snapshots only (bits $b_i = (s_i + 1)/2$, Bernoulli decoder), $d = 8$, $\beta \in \{1, 4\}$
2. keep the **active** coordinates: $\overline{\mathrm{KL}}_k = \left\langle \frac{1}{2}(\mu_k^2 + \sigma_k^2 - \ln \sigma_k^2 - 1) \right\rangle_{\mathbf{s}} > 0.1$ nats on held-out snapshots
3. **probe**: latent mean $\mu_k(\mathbf{s})$ of every held-out snapshot, Pearson correlation with an observable $O(\mathbf{s})$,

$$\rho_{kO} = \frac{\langle \mu_k O \rangle - \langle \mu_k \rangle \langle O \rangle}{\sqrt{\mathrm{Var}\, \mu_k \; \mathrm{Var}\, O}} \in [-1, 1]$$

   ± 1: exact linear relation, 0: no linear relation
4. table (coordinates $\times$ observables) and scatter plots of μ_k against the best-matching O

Observables Computed from the Spins

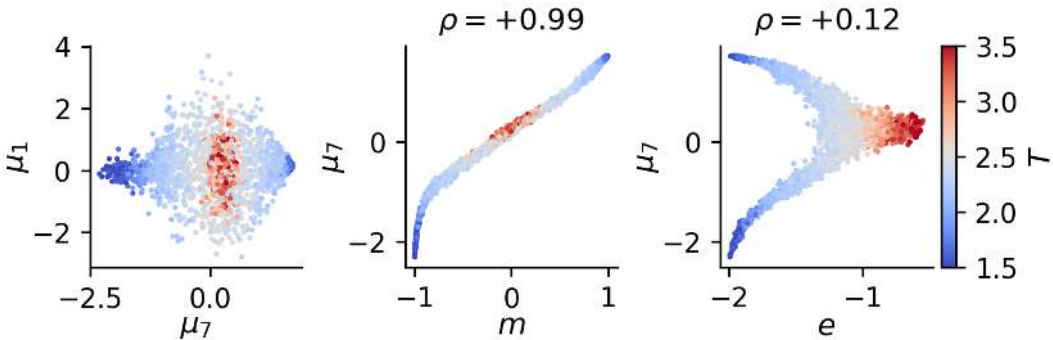
$N = L^2$ spins $s_{x,y} = \pm 1$:

- magnetisation $m = \frac{1}{N} \sum_{x,y} s_{x,y}$ and $|m|$: ferromagnetic order
- energy per spin $e = H/N$: satisfied and broken bonds
- stripe amplitudes $m_x = \frac{1}{N} \sum_{x,y} (-1)^x s_{x,y}$, $m_y = \frac{1}{N} \sum_{x,y} (-1)^y s_{x,y}$: spins alternating along x or y; $|m_x|$, $|m_y|$, $m_{\mathrm{str}} = \sqrt{m_x^2 + m_y^2}$
- simulation parameters: temperature T, coupling ratio $g = J_2/J_1$

ρ detects *linear* relations: a coordinate $\mu_k \propto m$ gives $|\rho| \approx 1$ with m but $\rho \approx 0$ with $|m|$; a symmetric relation such as $\mu_k \approx m^2$ needs $|m|$ or $|\mu_k|$.

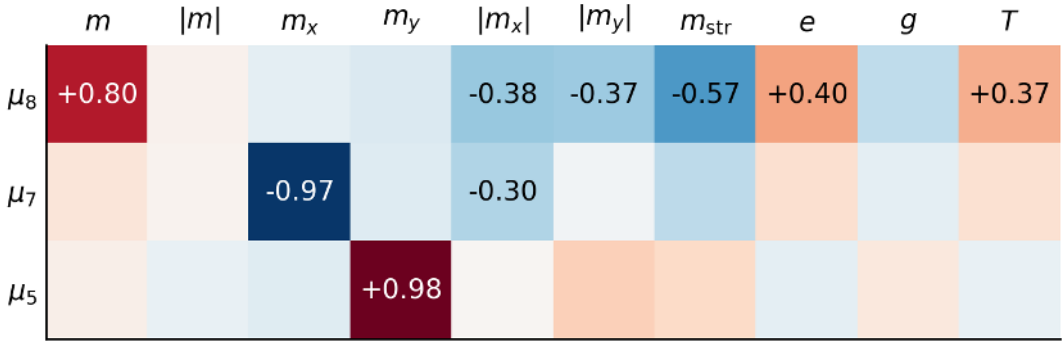
β below is always the KL weight, not $1/T$.

One Coordinate Is the Magnetisation



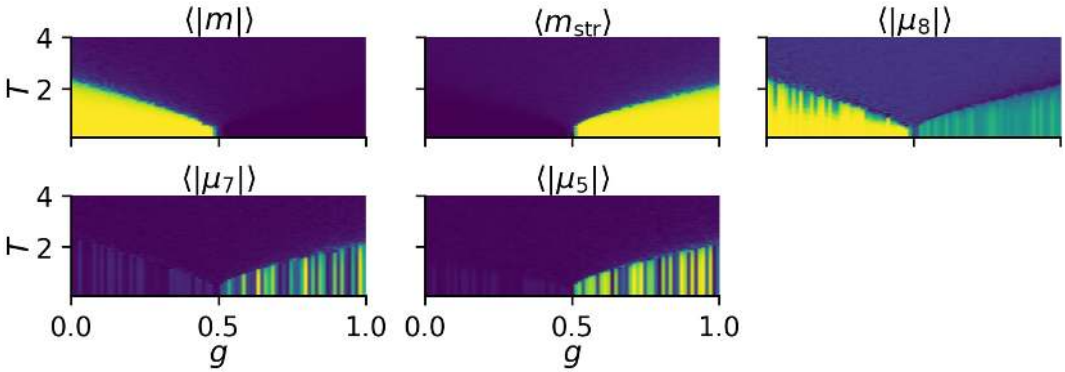
2D Ising, $L = 30$, $T \in [1.5, 3.5]$, 9472 snapshots (20% held out, shown, colour $= T$); $\beta = 4$, $d = 8$. Left: two most active coordinates; middle, right: the coordinate with the largest $|\rho|$ with m, with e. $\rho(\mu_7, m) = +0.99$ ($\beta = 1$ and 4); all other coordinates $|\rho| \leq 0.16$ with $m, |m|, e, T$. Ordered phases at the two ends of the μ_7 axis, disordered phase at its centre; e is related to $|\mu_7|$ ($\rho(|\mu_7|, e) = -0.88$), invisible to a linear probe.

J_1–J_2: Three Coordinates Are m, m_x and m_y



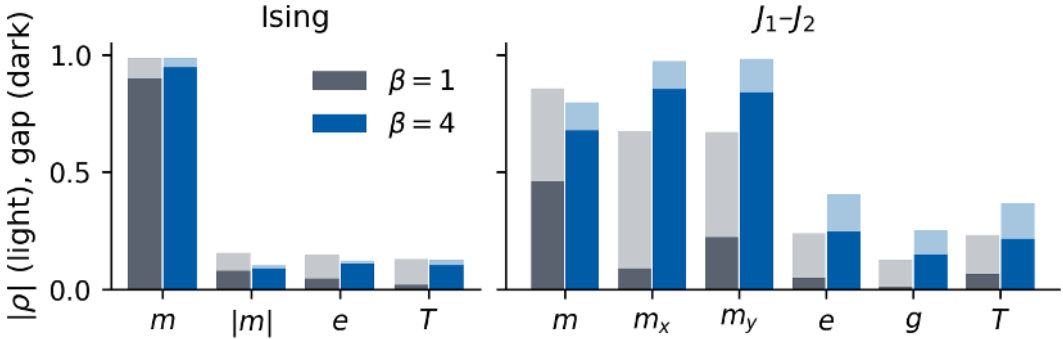
J_1–J_2, $L = 32$, 81×79 grid of (g, T), 12 snapshots per point (10 training, 2 held out); $\beta = 4$; rows: active coordinates, colour and numbers: $\rho(\mu_k, O)$ on held-out snapshots (numbers if $|\rho| \geq 0.3$). $\rho(\mu_7, m_x) = -0.97$, $\rho(\mu_5, m_y) = +0.98$, $\rho(\mu_8, m) = +0.80$, found without g, T or H. With $\beta = 1$: m in one coordinate (-0.86), m_x, m_y mixed over two (a rotated stripe plane).

The Coordinates Map the Phase Diagram



Over (g, T): mean over 12 snapshots of $|m|$, m_{str}, $|\mu_k|$ ($\beta = 4$); colour: min (dark) to max (yellow). $|\mu_8|$: ferromagnet, with some leakage into the stripe phase; $|\mu_7|$, $|\mu_5|$: stripe phase, in complementary bands (an artefact of the annealed chains; in equilibrium both stripe directions are equally likely); phase boundaries as in $\langle|m|\rangle$, $\langle m_{\mathrm{str}}\rangle$.

Larger β Gives Each Order Parameter Its Own Coordinate



For each observable O, over the active coordinates: light bar $|\rho|_{(1)} = \max_k |\rho_{kO}|$; dark bar: gap $|\rho|_{(1)} - |\rho|_{(2)}$ to the second largest (the MIG idea with correlations): large gap = one coordinate, and only one, follows O. Ising: one order parameter, gap 0.90 ($\beta = 1$) and 0.95 ($\beta = 4$). J_1–J_2: gap for m_x 0.09 $\to$ 0.85, m_y 0.22 $\to$ 0.84, m 0.46 $\to$ 0.68; e, g, T: no linear match ($|\rho| \leq 0.40$).

From a Check to a Discovery Tool

On an unknown model, an active coordinate whose $\langle|\mu_k|\rangle$ changes sharply across a parameter range is a *candidate order parameter*. Next step: express it through the spins.

Example on J_1–J_2, as if the answer were unknown: least-squares fit of μ_k by 8 standardised features $m, m_x, m_y, |m|, |m_x|, |m_y|, c_{\mathrm{nn}}, c_{\mathrm{nnn}}$ ($c_{\mathrm{nn}} = \frac{1}{4N}\sum_i s_i \sum_{j\in \mathrm{nn}(i)} s_j$, c_{nnn} the same for diagonal neighbours):

$$R^2 = 1 - \frac{\sum_{\mathbf{s}} (\mu_k - \hat{\mu}_k)^2}{\sum_{\mathbf{s}} (\mu_k - \langle \mu_k \rangle)^2} \quad \text{(fraction of the coordinate explained by the features)}$$

$\beta = 4$ model: $R^2 = 0.982$ for z_8 (largest coefficient m, then $-|m_x|$), 0.992 for z_7 (m_x), 0.989 for z_5 (m_y).

- the largest coefficient names the coordinate; a low R^2 means a quantity outside our library
- richer libraries and symbolic regression turn a coordinate into a formula

What a Probe Cannot Tell

- correlation **confirms, does not explain**: m, e and T are themselves correlated across a transition
- a coordinate may **mix** quantities; sign and scale of a coordinate are arbitrary
- a **nonlinear** relation ($\mu_k \approx m^2$) escapes a linear probe of m
- which coordinates appear depends on the model, the data, the architecture, β and the seed (Locatello et al.)
- **we only find what we probe for**: a coordinate matching none of our observables is noise or a quantity we have not thought of

Examples in Lab 026, held-out snapshots

Ising, $\beta = 4$: $\rho(|\mu_7|, \cdot) = +0.92$ ($|m|$), -0.88 (e), -0.78 (T); linear: $\rho(\mu_7, m) = +0.99$, $\rho(\mu_7, |m|) = -0.10$. J_1–J_2, $\beta = 1$: μ_5 mixes m_x (-0.58) and m_y ($+0.67$).

Conditioning and Disentangling Answer Different Questions

	conditional VAE (Labs 024, 025)	β-**VAE** (Lab 026)
training data	pairs $(\mathbf{x}, \mathbf{c})$	images $\mathbf{x}$ only
loss	$D + R$, $\mathbf{c}$ into encoder and decoder	$D + \beta R$
factor	chosen by us, given as $\mathbf{c}$	found by the network, named afterwards
guarantee	$\mathbf{c}$ controls what it labels	none in general
$\mathbf{z}$ carries	all but $\mathbf{c}$	all, ideally one factor per coordinate
sampling	fix $\mathbf{c}$, draw $\mathbf{z}$; or one $\mathbf{z}$, many $\mathbf{c}$	draw $\mathbf{z}$; move one found z_k
question	"generate this"	"what varies in these data?"

Complements: factors found by a β-VAE can become labels of a cVAE; physics: condition on (g, T) to generate, an unconditioned β-VAE to find collective variables.

9

Physical Parameters from Trajectories



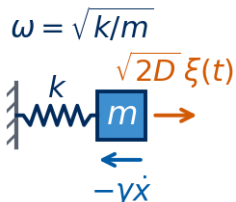
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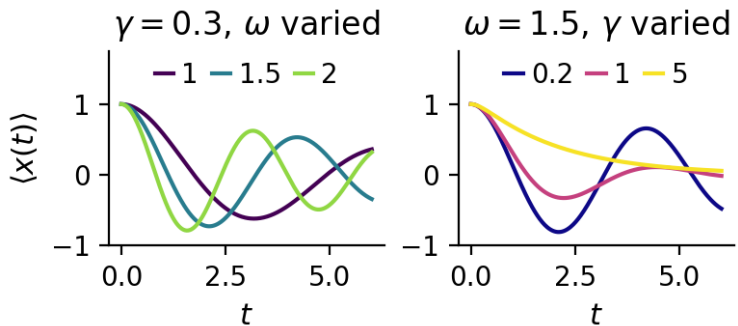
Frequency and Damping Shape a Noisy Oscillation

$$\ddot{x} = -\omega^2 x - \gamma\,\dot{x} + \sqrt{2D}\,\xi(t), \qquad \langle \xi(t) \rangle = 0, \quad \langle \xi(t)\,\xi(t') \rangle = \delta(t - t')$$

$\omega = \sqrt{k/m}$: frequency without friction; γ: friction rate; ξ: Gaussian white noise (independent kicks); D: noise strength ($= \gamma k_B T/m$ for a heat bath)

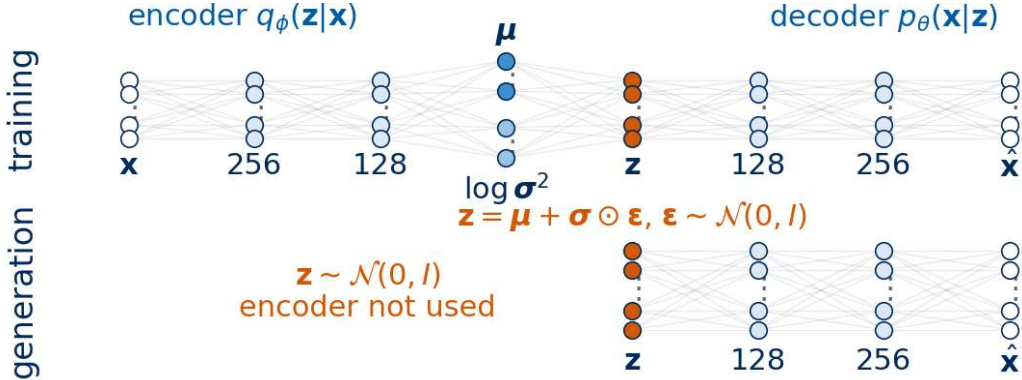
underdamped $\gamma < 2\omega$: oscillation at $\omega_d = \sqrt{\omega^2 - \gamma^2/4}$ inside $e^{-\gamma t/2}$; **overdamped** $\gamma > 2\omega$: creep back at the slow rate $\approx \omega^2/\gamma$



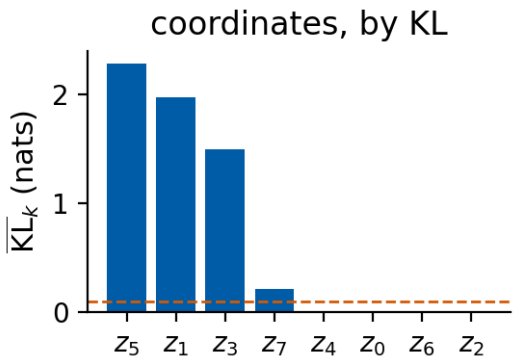


A β-VAE Compresses Each Trajectory

Gaussian decoder $p_\theta(\mathbf{x}|\mathbf{z}) = \prod_i \mathcal{N}(x_i;\, G_\theta(\mathbf{z})_i,\, \sigma_x^2) \Rightarrow$
$\mathcal{L}_\beta = \mathbb{E}_{q_\phi}\left[\|\mathbf{x} - G_\theta(\mathbf{z})\|^2/(2\sigma_x^2)\right] + \beta\,\mathrm{KL}(q_\phi(\mathbf{z}|\mathbf{x}) \,\|\, \mathcal{N}(0, I))$
reconstruction = mean squared error $\times\, 100/(2\sigma_x^2)$, $\sigma_x = 0.1$; G_θ: mean trajectory; $d = 8$;
$\beta \in \{0.5, 1, 4, 8\}$, 3 seeds each



Damping Gets Its Own Coordinate, Frequency Does Not



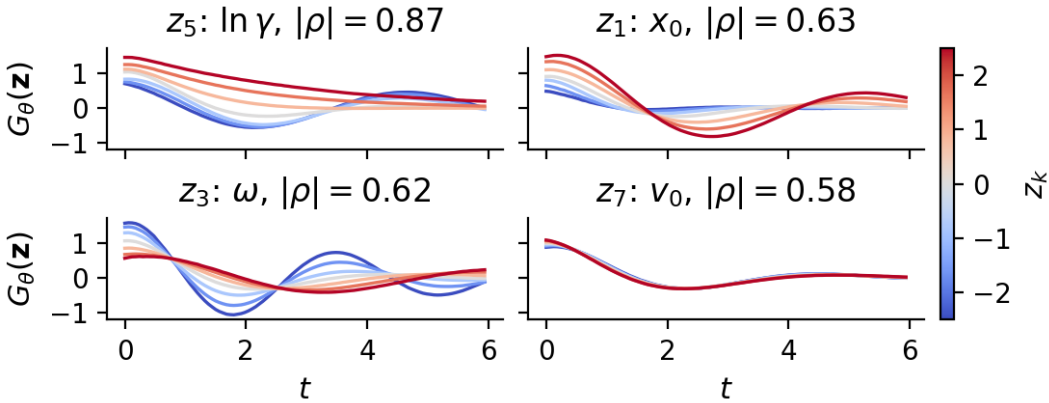
$|\rho(\mu_k, \text{factor})|$

	ω	$\ln\gamma$	x_0	v_0
z_5	0.27	0.87	0.19	0.18
z_1	0.40	0.30	0.63	0.26
z_3	0.62	0.11	0.46	0.11
z_7	0.08	0.16	0.23	0.58

$\beta = 4$, seed 2 (largest MIG of three seeds), 5000 test trajectories; ρ: Pearson correlation of latent mean μ_k and factor; active: $\overline{\mathrm{KL}}_k > 0.1$ nats (dashed)

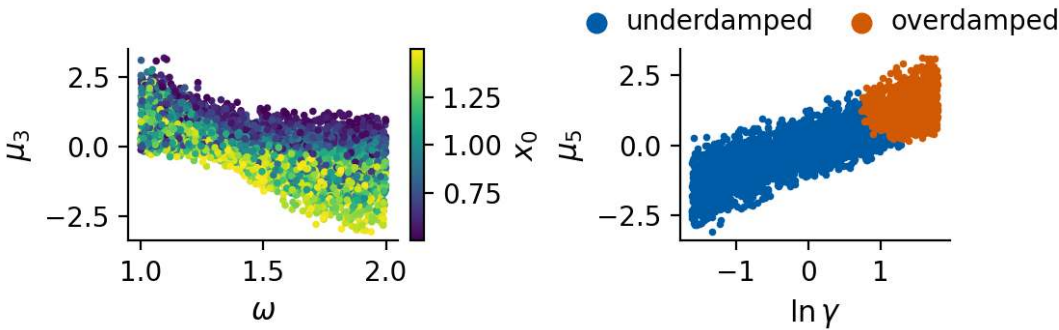
4 active coordinates for 4 factors; damping $|\rho| = 0.87$, clean; ω (0.62) and x_0 (0.63) shared by two coordinates

Traversals Separate Damping from Frequency



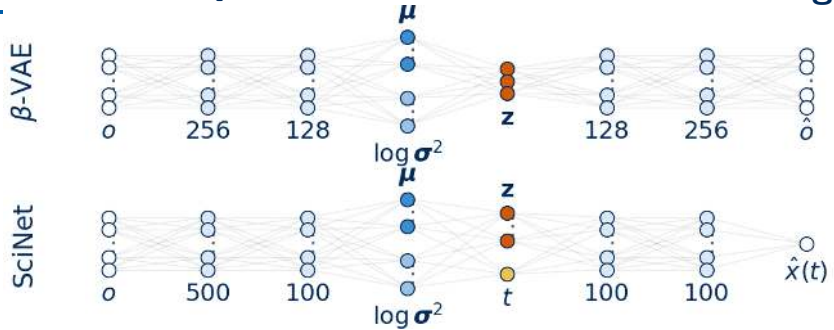
$\boldsymbol{\mu}$ of one test trajectory ($\omega \approx 1.5$, $\gamma \approx 1$, $x_0 \approx 1$, $v_0 \approx 0$); one z_k from -2.5 to 2.5, the others fixed; title: factor with the largest $|\rho|$. The ω-coordinate also moves $x(0)$.

Overdamped: the Damping Coordinate Measures ω^2/γ



- right: $|\rho(\mu_\gamma, \ln\gamma)| = 0.80$ underdamped, 0.13 overdamped, where $|\rho(\mu_\gamma, \ln\frac{\omega^2}{\gamma})| = 0.69$: the slow rate is what the data show
- left: slope of μ_ω grows with x_0; R^2 of a fit: ω 0.38; $\omega, x_0, \omega x_0$ 0.63

SciNet Answers Questions Instead of Reconstructing



- Iten et al. (PRL 2020): encoder compresses trajectory; decoder answers a question (predicts $x(t)$ at time t)
- cost: prediction error $+ \beta\,\mathrm{KL}$ ($\beta = 10^{-3}$, 3 latent neurons)
- physical parameters (κ, b) separate into independent neurons; standard β-VAE achieves the same with a stronger KL weight

R. Iten et al., Phys. Rev. Lett. **124**, 010508 (2020), doi:10.1103/PhysRevLett.124.010508

Trajectories Alone Reveal How Many Parameters Matter

- a β-VAE trained on trajectories alone finds *how many* numbers describe them and isolates the most visible parameter (damping); weakly visible or mutually confusable ones stay mixed
- check against physics before naming a coordinate: ω^2/γ in the overdamped regime is a correct answer, not a failure
- SciNet asks questions about the system instead of reconstructing the input; isolates κ and b, as the β-VAE does with a stronger KL term

R. Iten et al., PRL **124**, 010508 (2020), doi:10.1103/PhysRevLett.124.010508

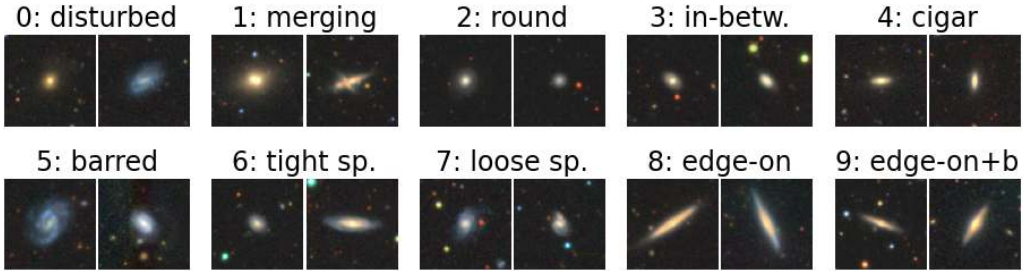
10

Galaxy Morphology with a β-VAE



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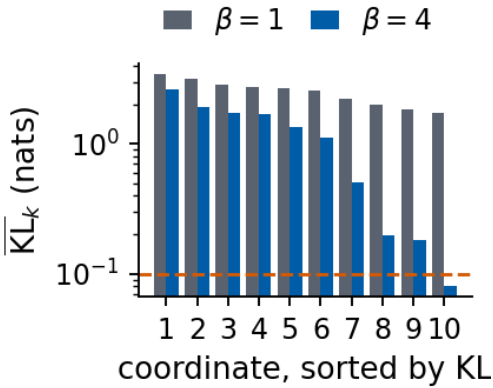
Galaxy10 DECaLS: 17 736 Galaxies in Ten Classes



- one sample: 64×64 pixels of $1.05''$; filter z (near infrared) $\rightarrow$ red, r (red) $\rightarrow$ green, g (blue-green) $\rightarrow$ blue; value: arcsinh-stretched brightness
- images: Dark Energy Camera Legacy Survey (DECaLS); classes: Galaxy Zoo votes

H. W. Leung, J. Bovy, Galaxy10 DECaLS, CC BY 4.0, Zenodo; M. Walmsley et al., MNRAS 509, 3966 (2022); A. Dey et al., AJ 157, 168 (2019)

Compressing Galaxy Images: Active Coordinates

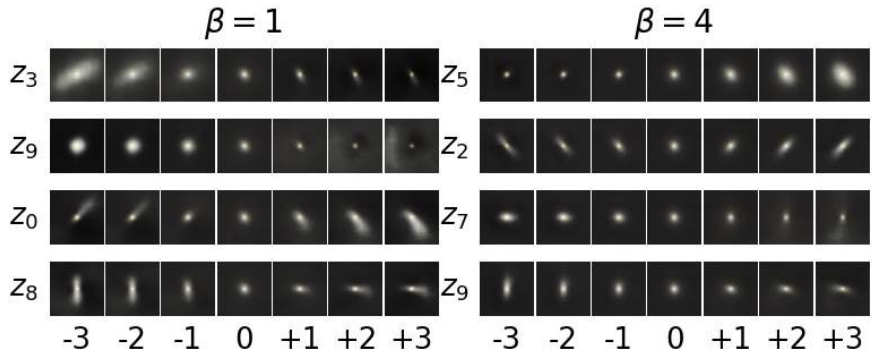


Continuous pixels $\Longrightarrow$ Gaussian decoder loss:

$$\mathcal{L}_\beta = \mathbb{E}_{q_\phi}\left[\frac{\|\mathbf{x} - \hat{\mathbf{x}}_\theta(\mathbf{z})\|^2}{2\sigma_x^2}\right] + \beta \sum_k \mathrm{KL}_k$$

- $\beta = 1 \to 4$: rate drops from 25.3 to 11.4 nats
- active coordinates cut from 10 to 9 (out of 16)
- $\beta = 4$ enforces strict bottleneck: only dominant macroscopic factors survive

Traversals Separate Size, Brightness and Shape



Row: $\hat{\mathbf{x}}_\theta(\mathbf{z})$, $z_k = \mu_k(\mathbf{x}_0) + v$, other coordinates fixed

- $\beta = 4$: z_5 size and brightness; z_2 diagonal stretch; z_7, z_9 stretch along the axes
- $\beta = 1$: size, brightness and shape change together; no row changes the colour

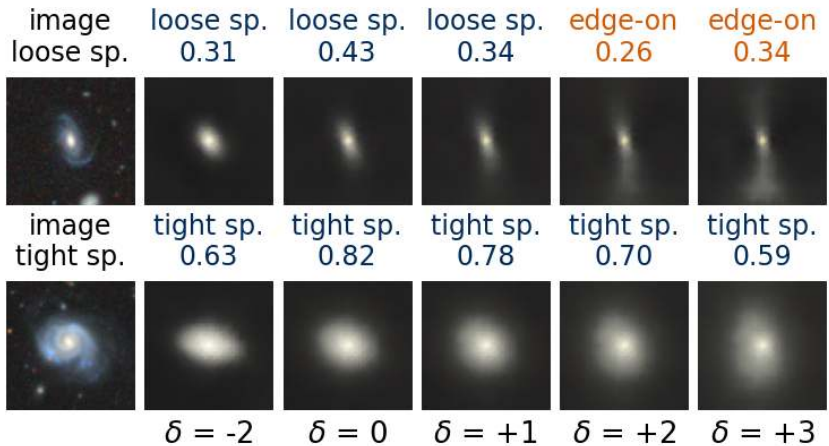
Latent Coordinates Recover Physical Quantities

	R	q	e_1	e_2	F	c	C
z_5	0.93	0.06	0.04	0.14	0.97	0.22	0.15
z_2 (e_2)	0.36	0.07	0.06	0.78	0.39	0.06	0.12
z_7 (e_1)	0.14	0.13	0.60	0.05	0.20	0.13	0.14
z_9 (e_1)	0.19	0.12	0.80	0.14	0.21	0.13	0.12
z_3	0.17	0.12	0.48	0.14	0.12	0.11	0.15
z_1	0.19	0.04	0.30	0.33	0.17	0.00	0.02
z_8	0.26	0.05	0.49	0.36	0.26	0.00	0.02
z_4	0.02	0.08	0.34	0.27	0.01	0.04	0.08
z_0 (e_2)	0.18	0.10	0.18	0.71	0.22	0.06	0.13

$\beta = 4$; Spearman $|\rho|$ of $\mu_k(\mathbf{x})$ vs physical quantity (2 736 test galaxies)

- **Size & brightness**: isolated in z_5 ($|\rho| = 0.93$ with radius R, 0.97 with flux F)
- **Orientation & ellipticity**: stored as two independent components (e_1, e_2) in z_7, z_9 and z_0, z_2
- **Axis ratio** q: nonlinear in (e_1, e_2), so $|\rho| \leq 0.22$ on linear probe
- **Colour**: ignored by the bottleneck ($|\rho| \leq 0.27$); reconstructions stay grey

Counterfactual Galaxies: Elongation Turns Spirals Edge-On



$\mathbf{z} = \boldsymbol{\mu}_\phi(\mathbf{x}) + \delta\, \mathbf{e}_7$, $\delta > 0$: smaller q; of 249 spirals 12% edge-on at $\delta = 3$, 6% at -3

Summary

1. VAE & Sampling

Latent model: $p_\theta(\mathbf{x}) = \int p(\mathbf{z})p_\theta(\mathbf{x}|\mathbf{z})d\mathbf{z}$

ELBO: Reconstruction (BCE/MSE) minus KL information penalty pulling codes onto $\mathcal{N}(0, I)$.

Generative: Smooth latent space enables interpolation and new samples, but decoder produces independent pixels given $\mathbf{z}$ (blur/noise).

2. Conditional Physics

Conditioning: $\mathbf{c}$ selects macroscopic state (T, class); $\mathbf{z}$ carries style / fluctuations.

Spin snapshots: cVAE captures phases, symmetry breaking, and order parameter maps across (g, T).

Limitation: Factorised Bernoulli decoder acts like a mean-field model; misses exact bond energy fluctuations $e^{-H/T}/Z$.

3. β-VAE Disentangling

Bottleneck: $\beta > 1$ heavily penalises channel capacity, pushing independent factors to Cartesian axes.

Physical discovery: Recovers m in Ising, (m_x, m_y) in J_1–J_2, damping γ and slow rate ω^2/γ in oscillators.

Reality check: Disentanglement is partial and sensitive to data/seeds; factors must be verified by physics probes.

Thank You!

Questions, project ideas, feedback: marcin.plodzien@uj.edu.pl

Website · GitHub · Google Scholar

